

Hibernating bears get the majority of their energy from stored fatty acids



Eat 20 hrs per day for 7 months sleeping !

Lehninger

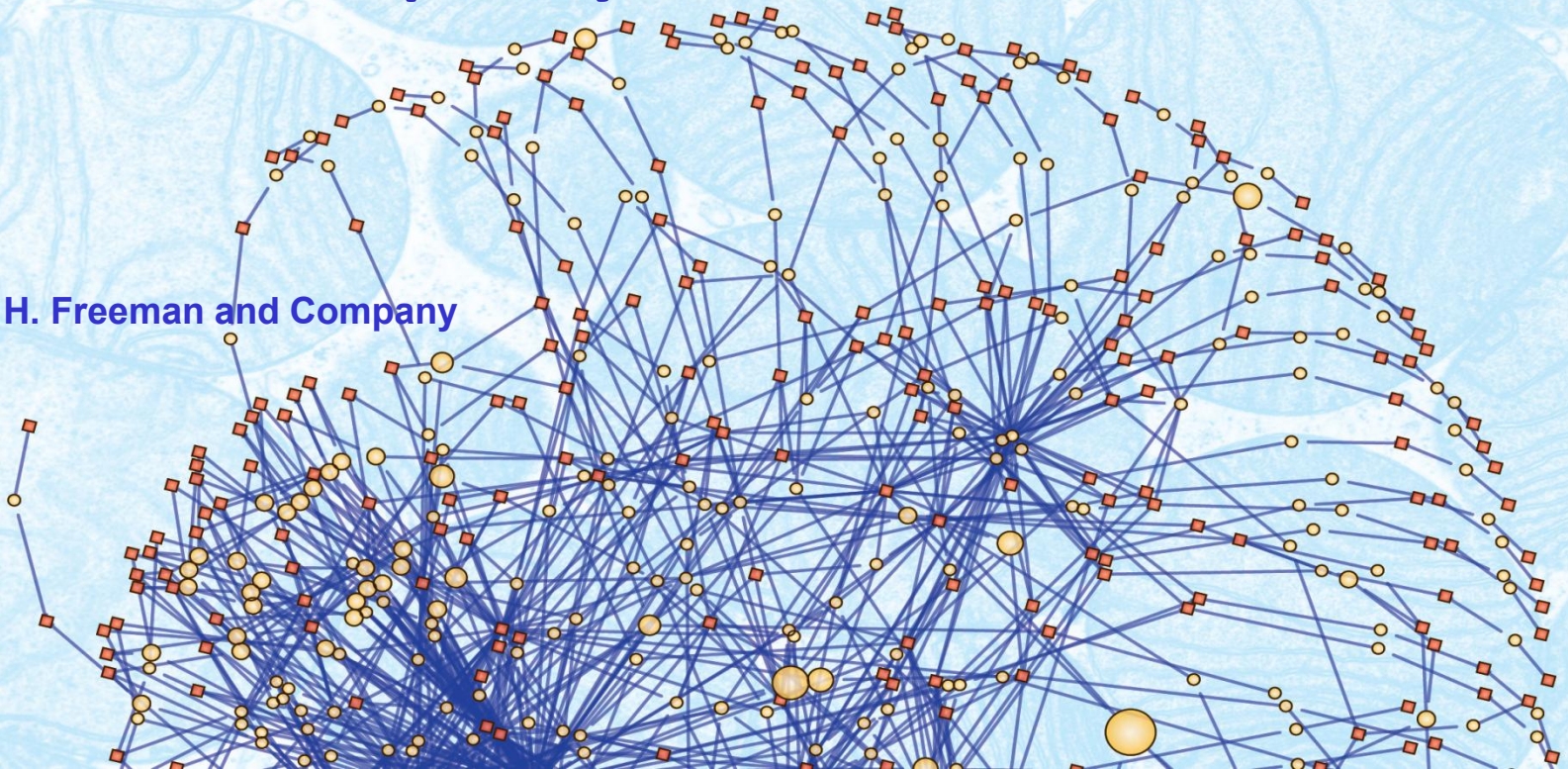
SIXTH EDITION

Principles of Biochemistry

David L. Nelson | Michael M. Cox

17 | Fatty Acid Catabolism

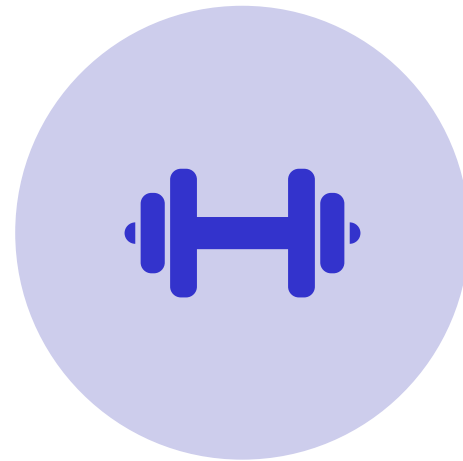
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Today, in this class..

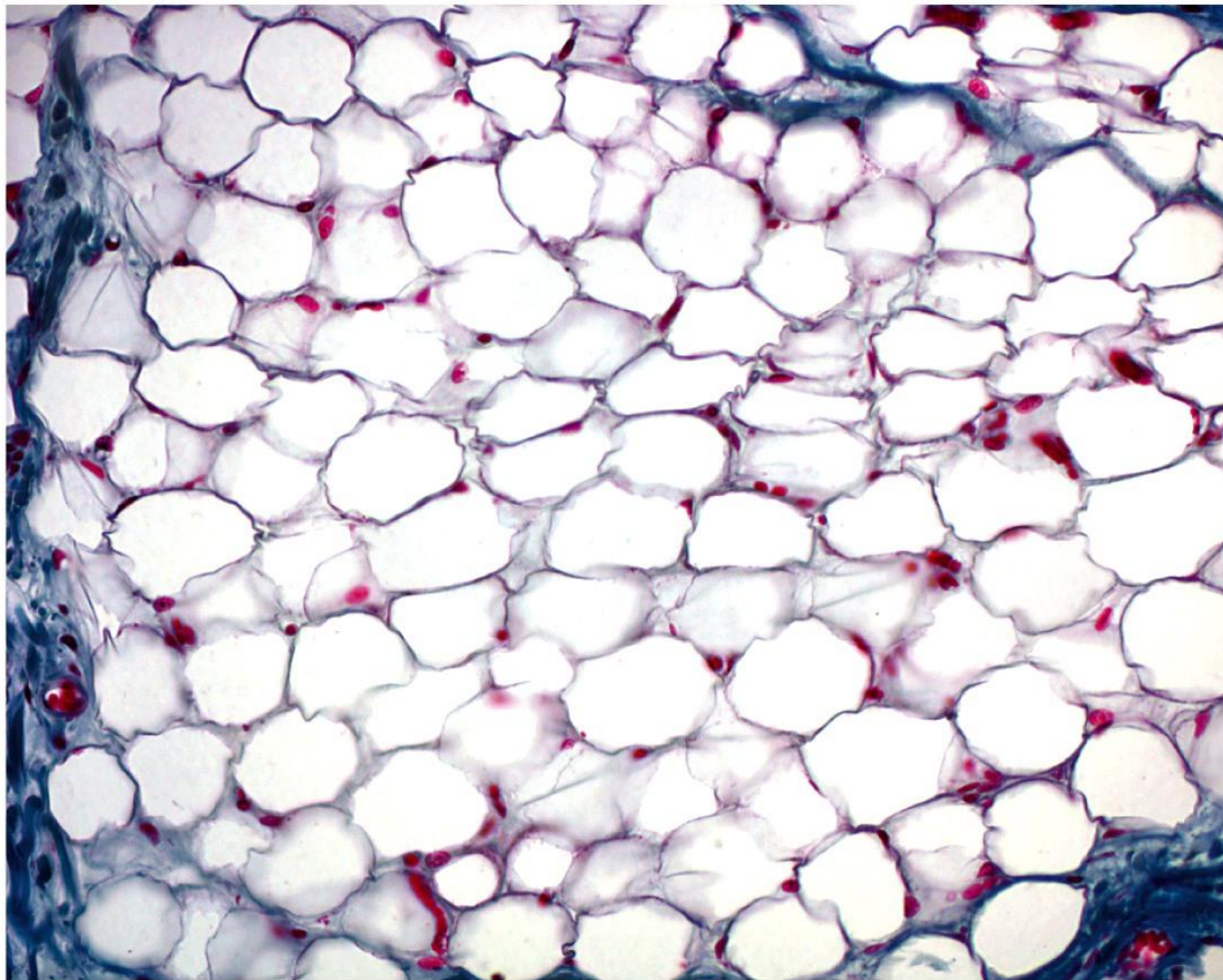


FAT STORAGE AND
MOBILIZATION



OXIDATION OF FATS

Fat Storage in White Adipose Tissue



125 μm

Figure 10-4a

Lehninger Principles of Biochemistry, Sixth Edition
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Fats provide efficient fuel storage

- The advantage of fats over polysaccharides:
 - Fatty acids carry more energy per carbon because they are more reduced
 - Fatty acids carry less water along because they are nonpolar
- Glucose and glycogen are for short-term energy needs, quick delivery
- Fats are for long-term (months) energy needs, good storage, slow delivery
- About 80% of energy needs of mammalian heart and liver are met by oxidation of fatty acids

17.1 Digestion, Mobilization, and Transport of Fats

- **Cell can obtain fatty acid fuels from:**

- ① **Diet**
- ② **Stored TG in cells**
- ③ **Synthesized FA**

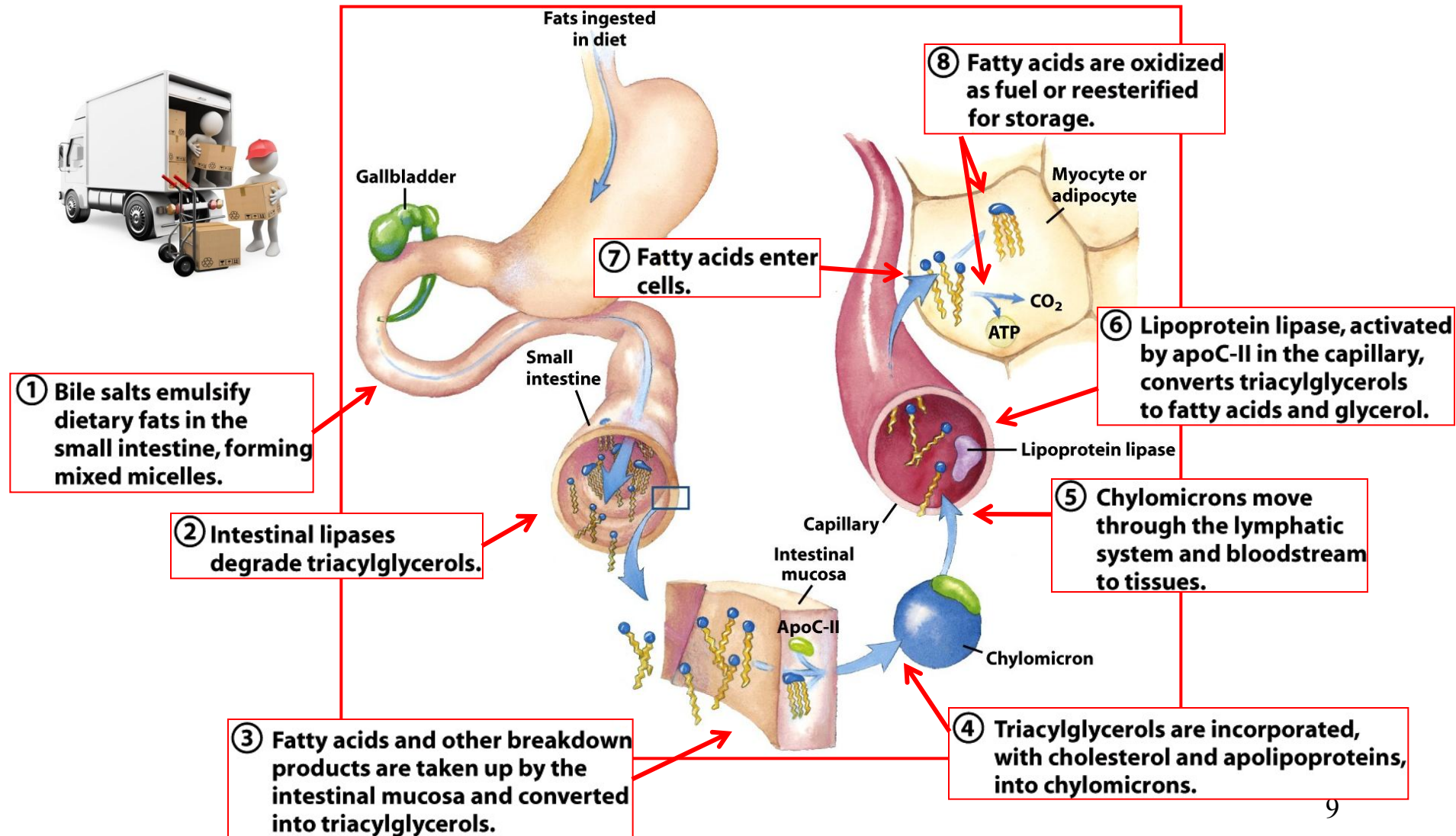


Dietary fats are absorbed and transported



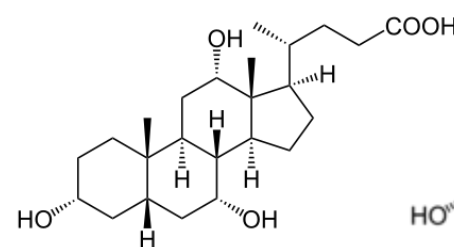
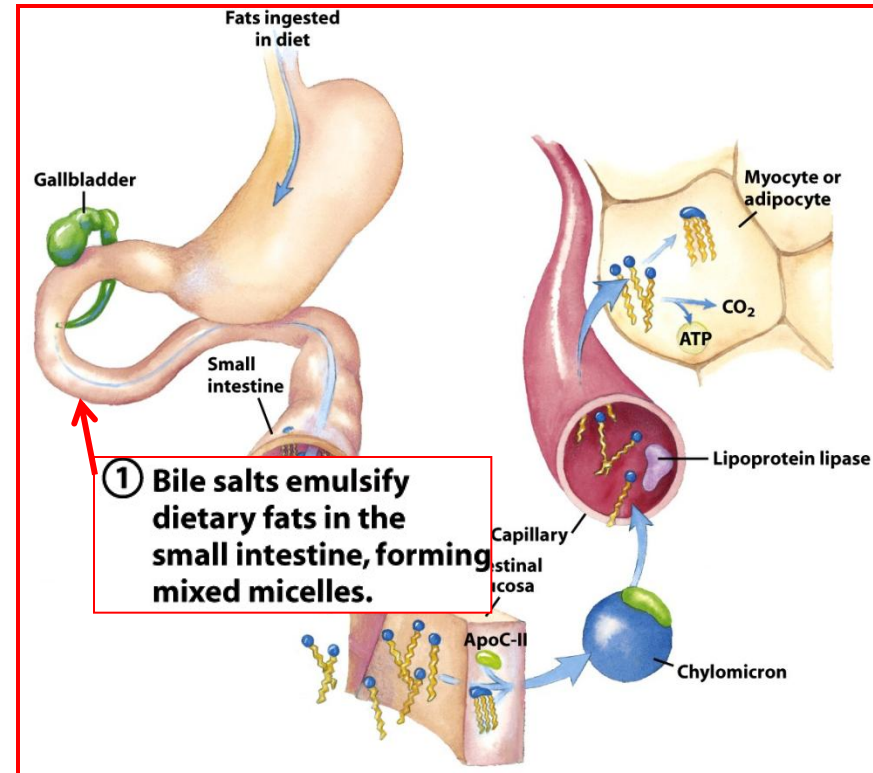
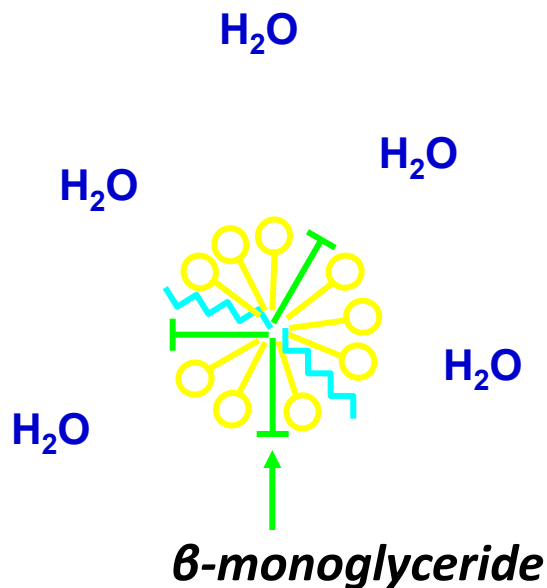
1) Dietary fats are absorbed in the small intestine

TG → FA → TG (Chylomicron) → FA → TG

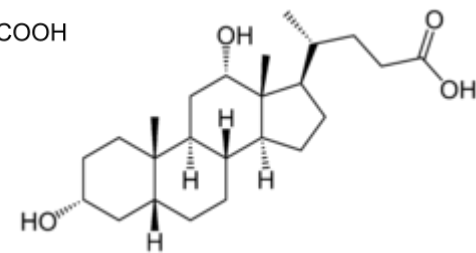


Step 1:

- **Bile salts: synthesized from cholesterol in liver, stored in gall bladder; biological detergent, mixed micelles (Bile S & TG)**



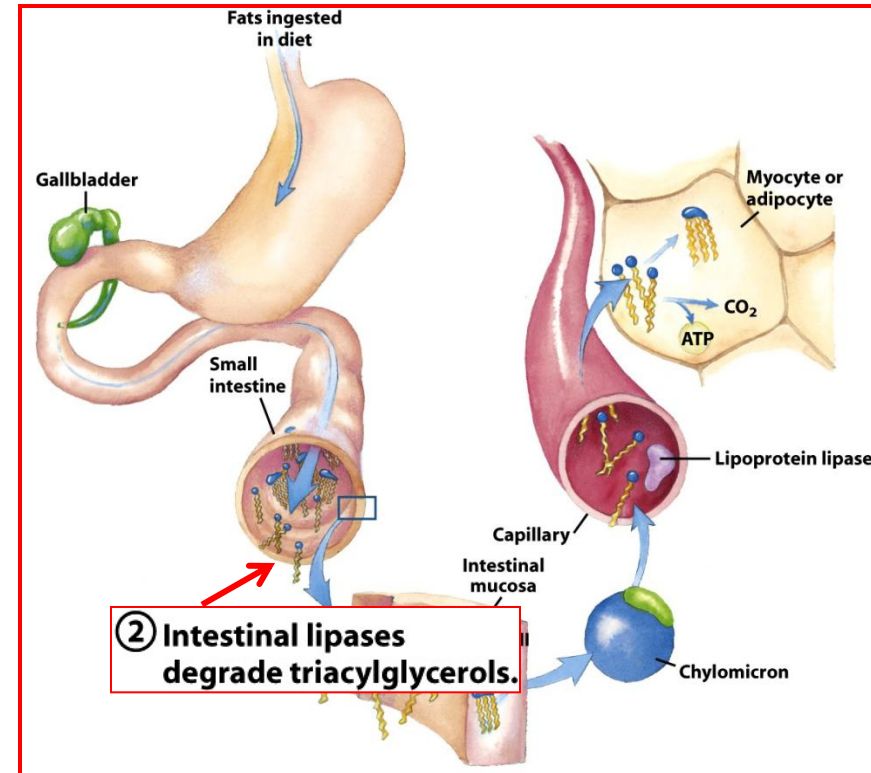
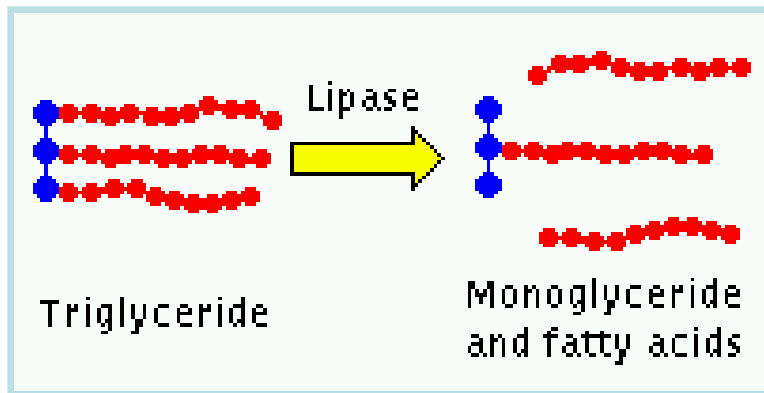
Cholic acid



Deoxycholic acid

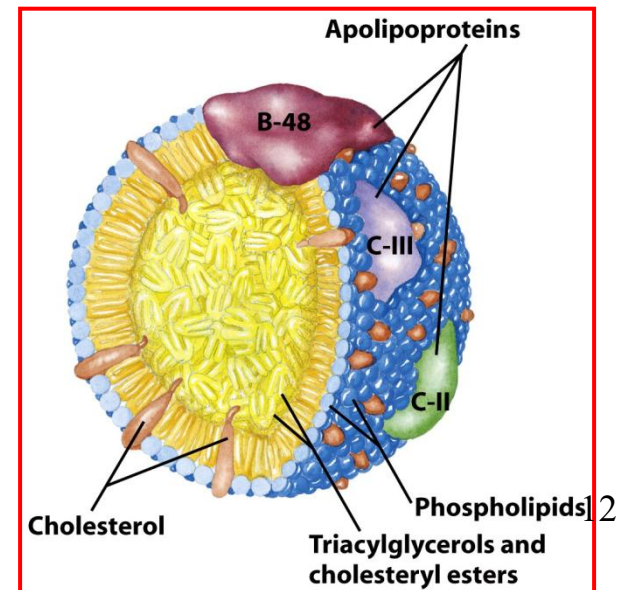
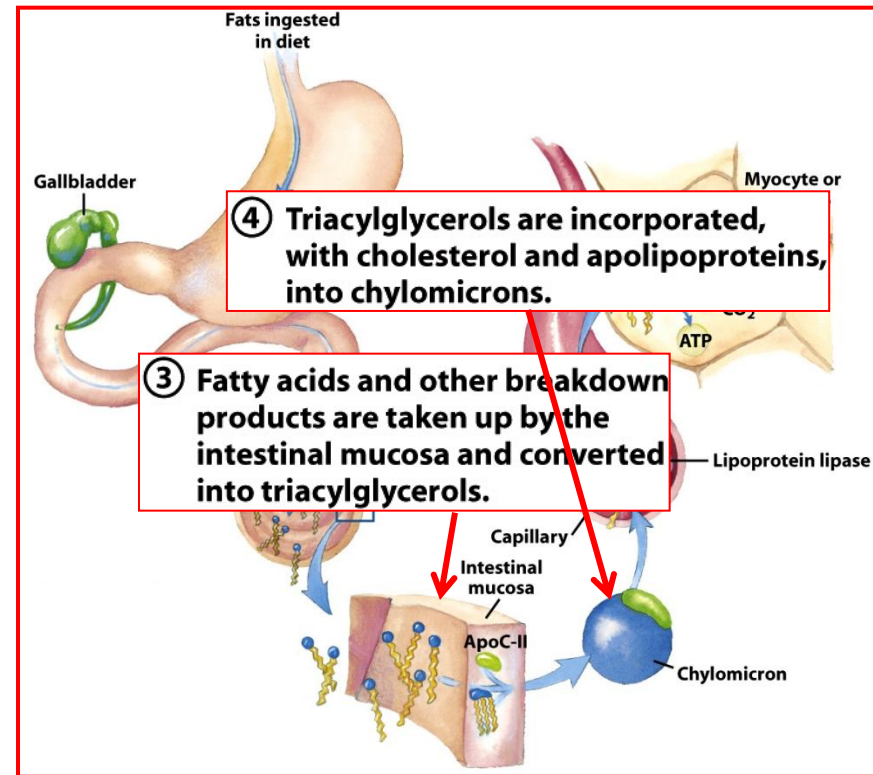
Step 2:

- **Lipases: Triacylglycerol (TG) to monoacylglycerol and diacylglycerol, free FA, and glycerol.**



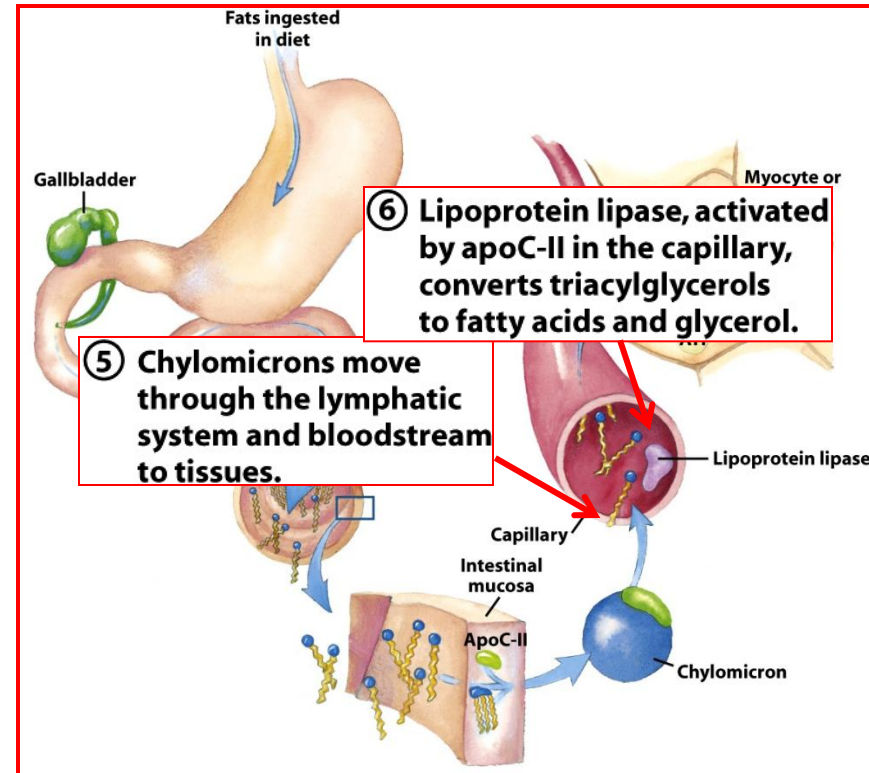
Step 3 & Step 4

- **Intestinal mucosa :**
Reconvert to TG
to form Chylomicron
- **Chylomicron:** packaged w/
dietary cholesterol and
apolipoproteins into
lipoprotein spherical
aggregates;
- **Apolipoprotein:** combine
with lipids to form
lipoprotein particles.



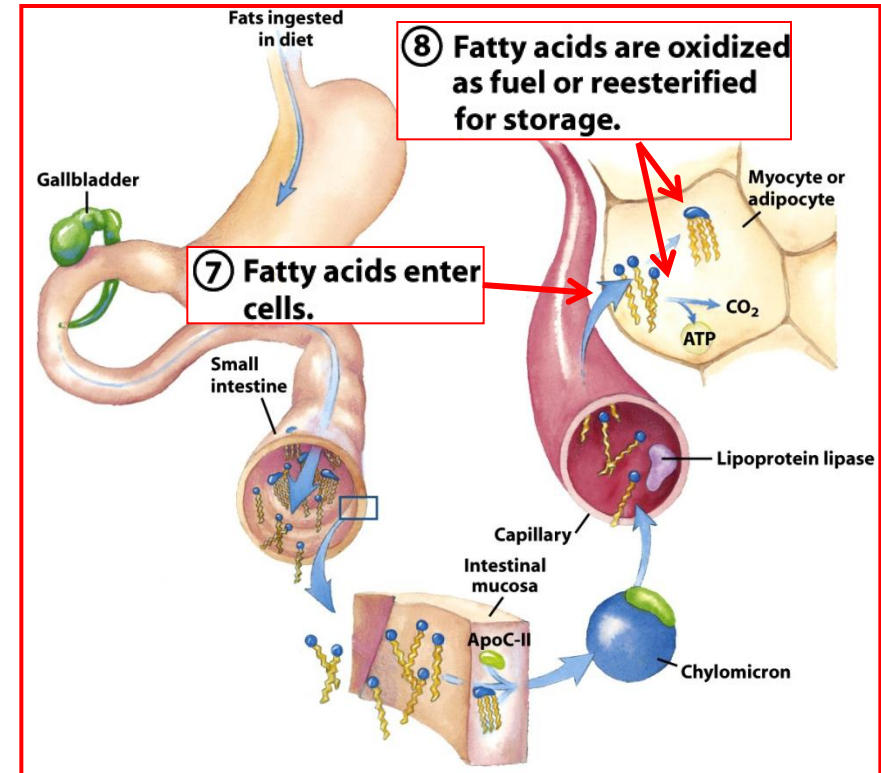
Step 5 & Step 6

- Chylomicron moves through lymphatic system & blood; **carried to muscle and adipose tissue**
- **Lipoprotein lipase**: convert TG back to FA's and glycerol to be taken by target cells.

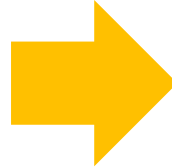


Step 7 & Step 8

- **FA's and glycerol are taken up by target tissues**
- **In muscle:** FA's are oxidized to produce energy
- **In adipose tissues:** reesterified for storage as TG

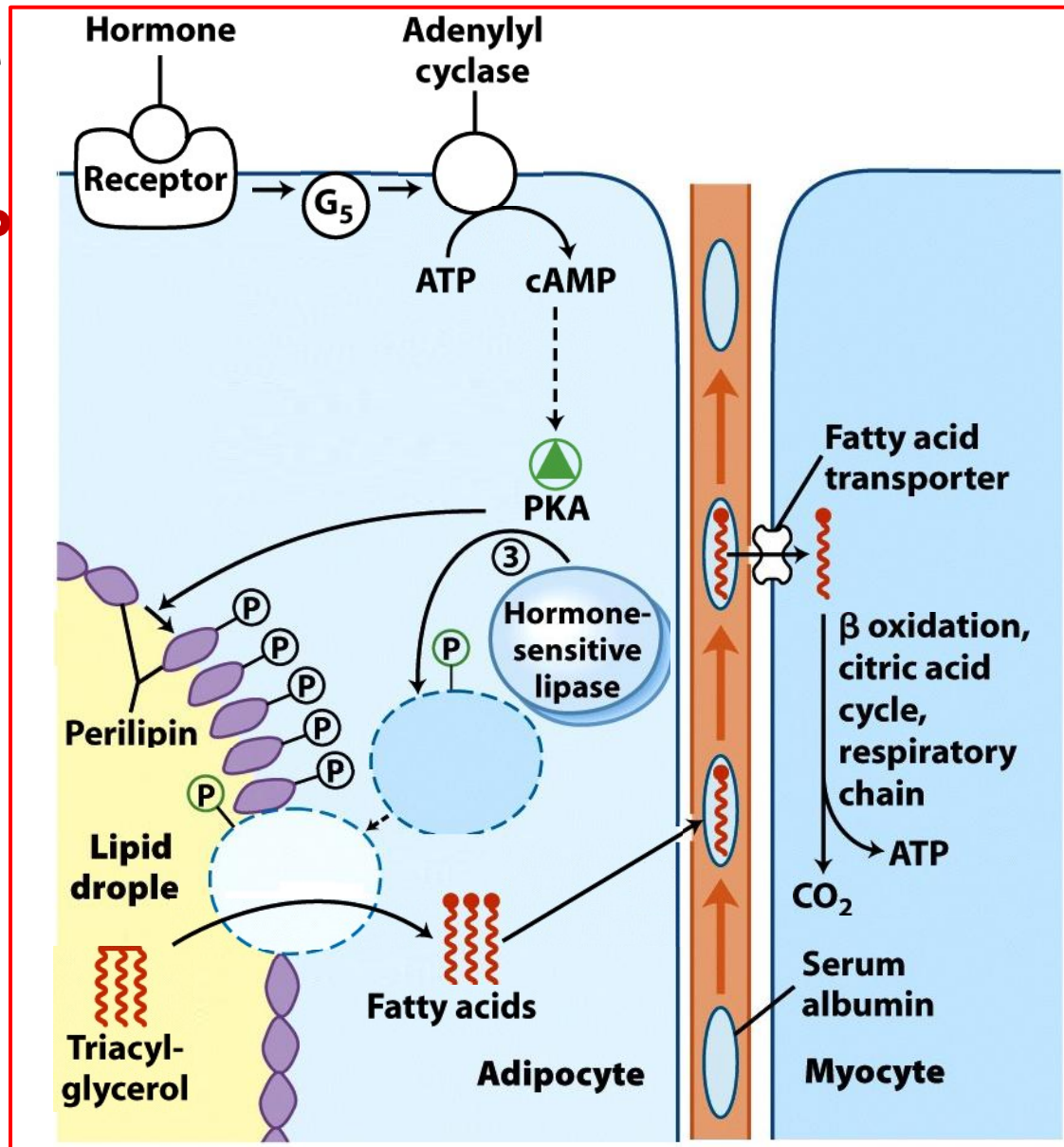


Mobilization of stored TG (triglyceride)



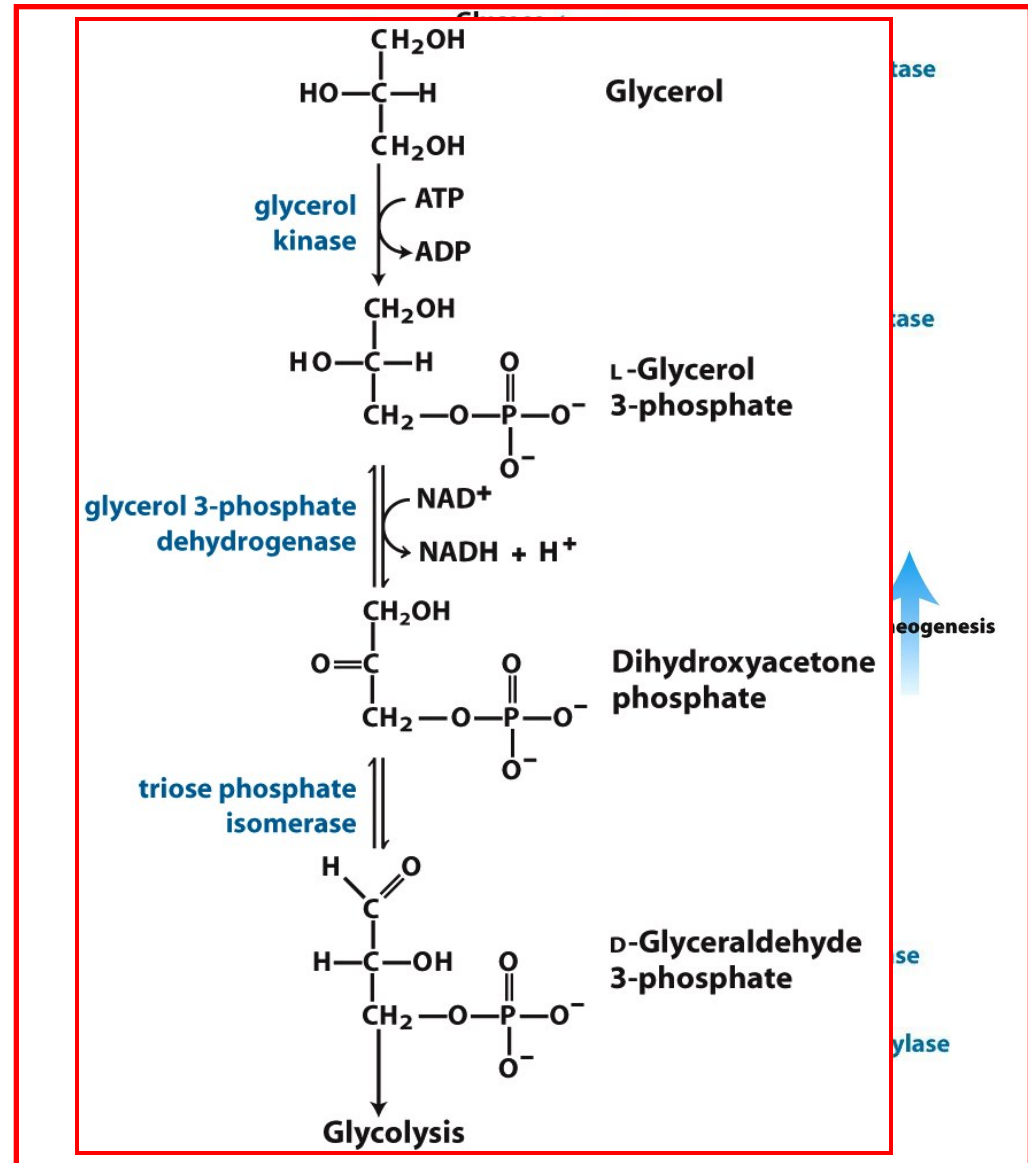
Mobilization of stored TG by Hormones (1)

- ① ↓[G]: glucagon, epine
- ② Adenylyl cyclase
→ **2nd messenger: cAMP**
→ **Activation of PKA**
- ③ H-S lipase: **P'ed by cAMP-depend PKA**
- ④ Perilipin: **P'ed**
- ⑤ TG to FA's by lipase
- ⑥ FA into blood: **serum albumin bound**
- ⑦ Moved to target cells
- ⑧ E production



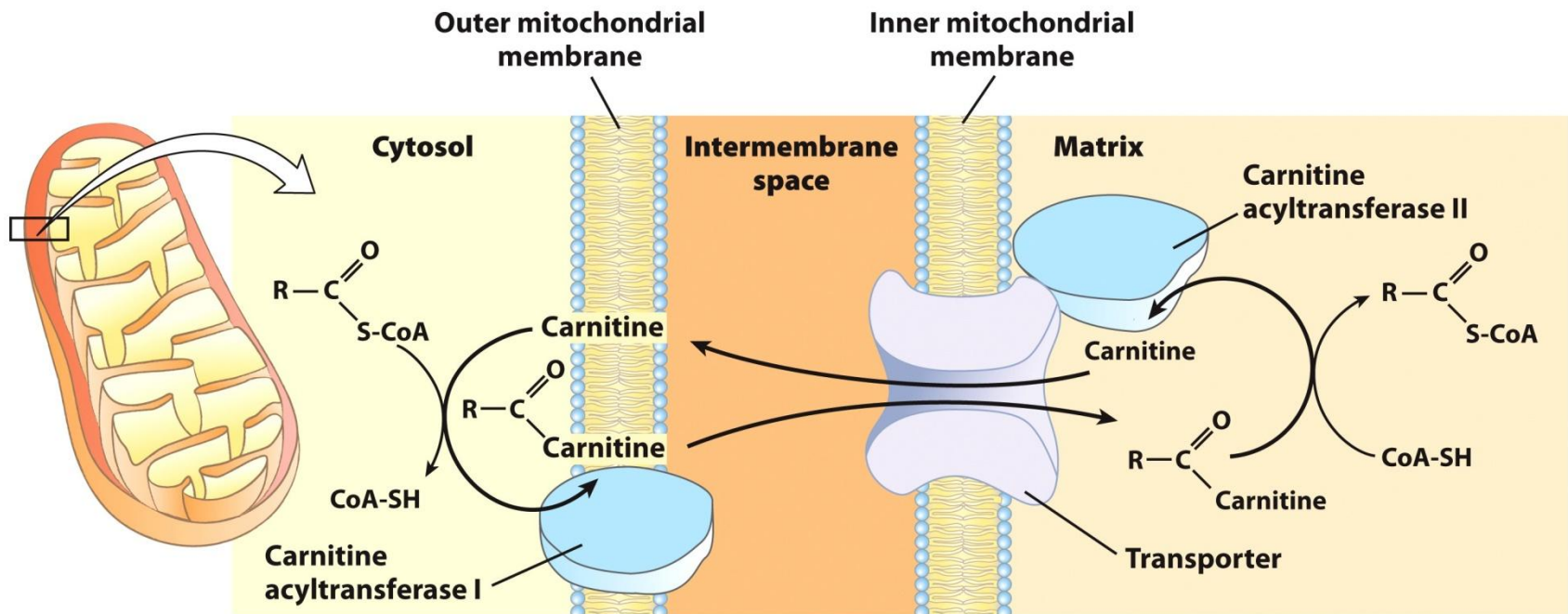
Mobilization of stored TG by Hormones (2)

- **Glycerol:**
phosphorylated by
glycerol kinase
- **Glyceraldehyde 3-P:**
oxidized by
glycolysis.



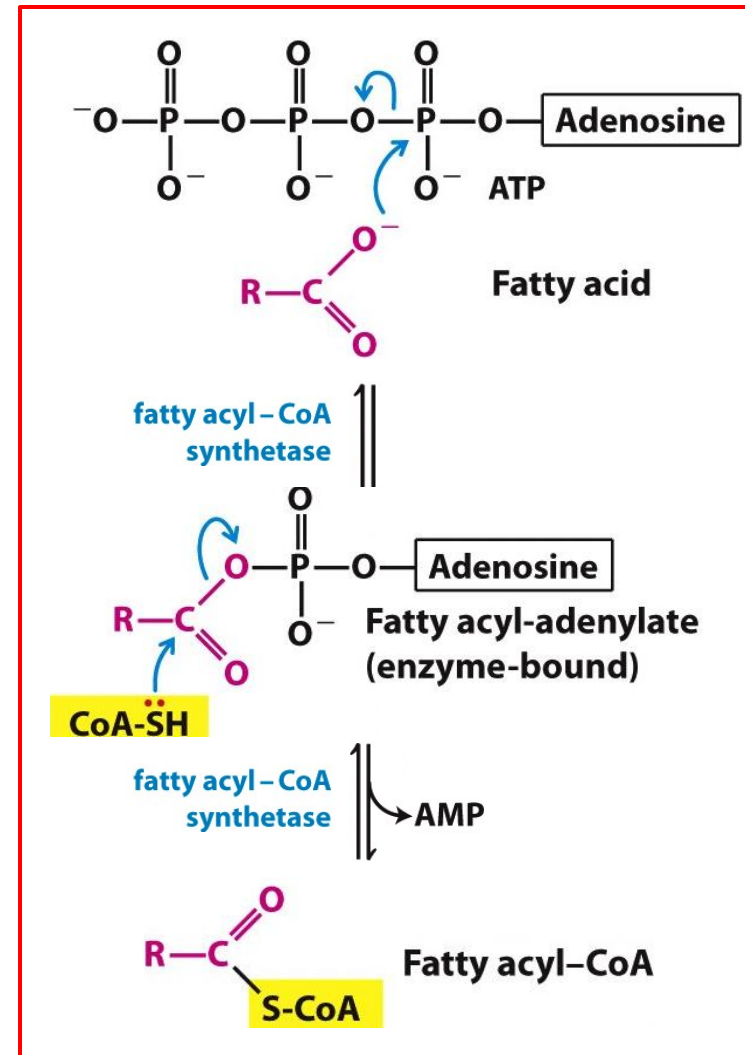
FA's are activated and transported into mitochondria (1) (carnitine shuttle)

- Enz's for FA oxidation: **in mitochondria**
- FA (~12 C): **enter mitochondria w/o help**
- FA (14~ C): **carnitine shuttle**



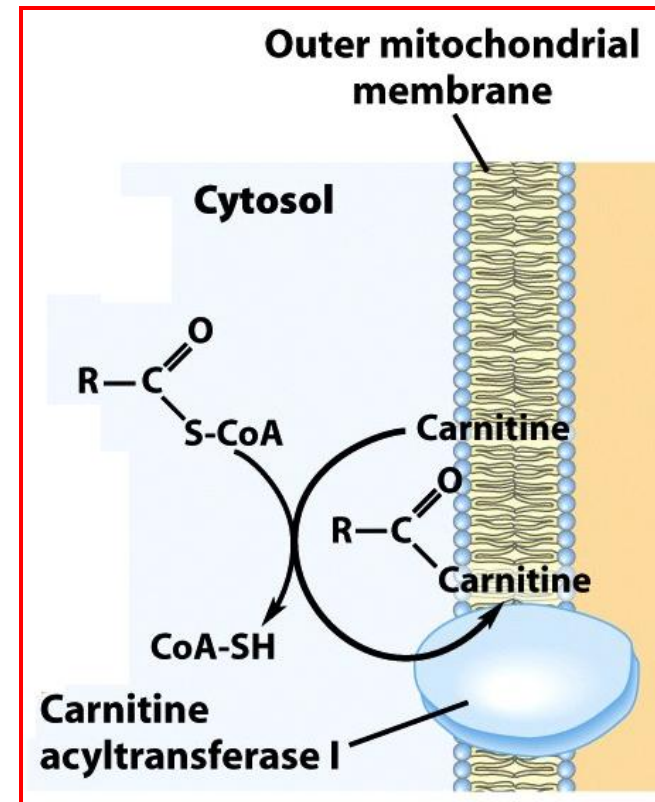
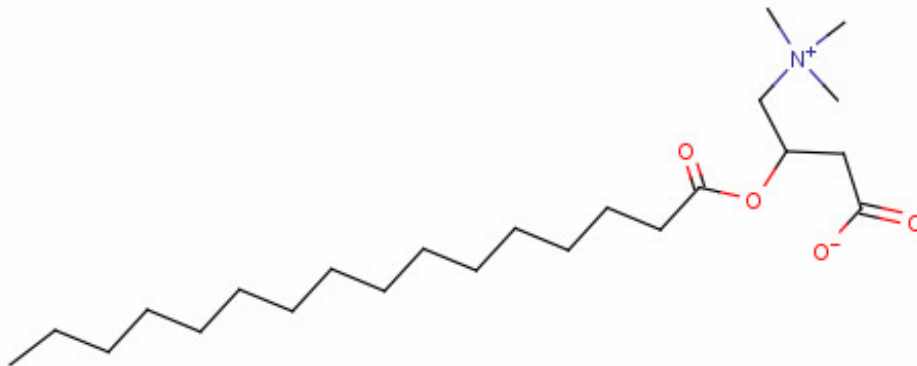
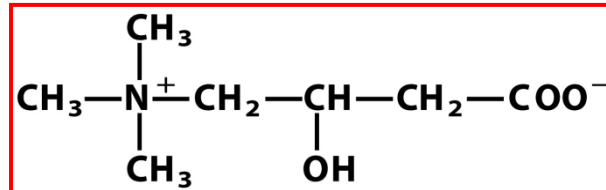
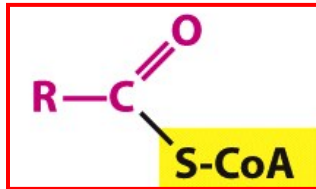
FA's are activated and transported into mitochondria (2)

- 1st rxn: **Acyl-CoA synthetase**
- **Formation of fatty acyl-CoA**
- **located in outer memb of mito**



FA's are activated and transported into mitochondria (3)

- 2nd rxn: **carnitine acyltransferase I** (located in outer memb of mito)
- Formation of fatty acyl-carnitine

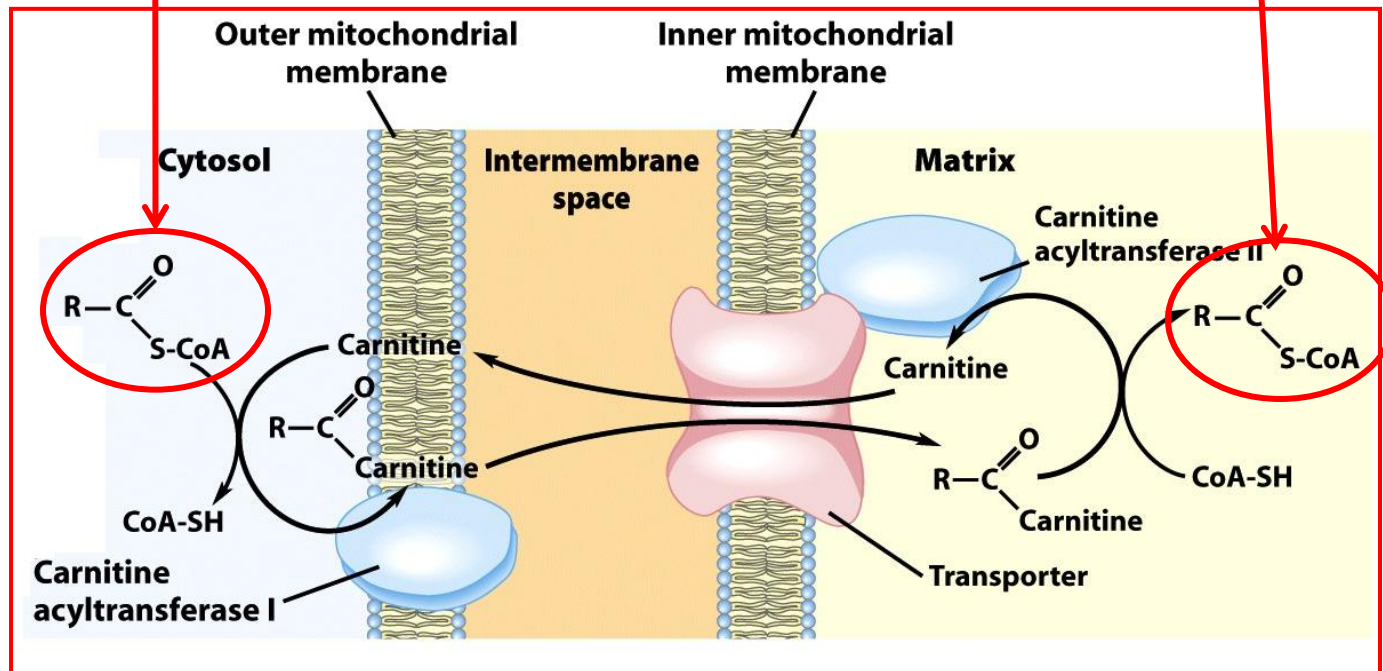


FA's are activated and transported into mitochondria (4)

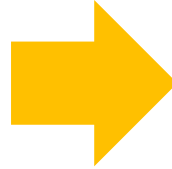
- 3rd rxn: **carnitine acyltransferase II** (located in inner memb of mito)
- **FA-Carnitine** moves into matrix thru transporter
- **FA-Carnitine** changed back to fatty acyl-CoA (w/ mitochondrial CoA) and carnitine

FA syn, memb lipid syn

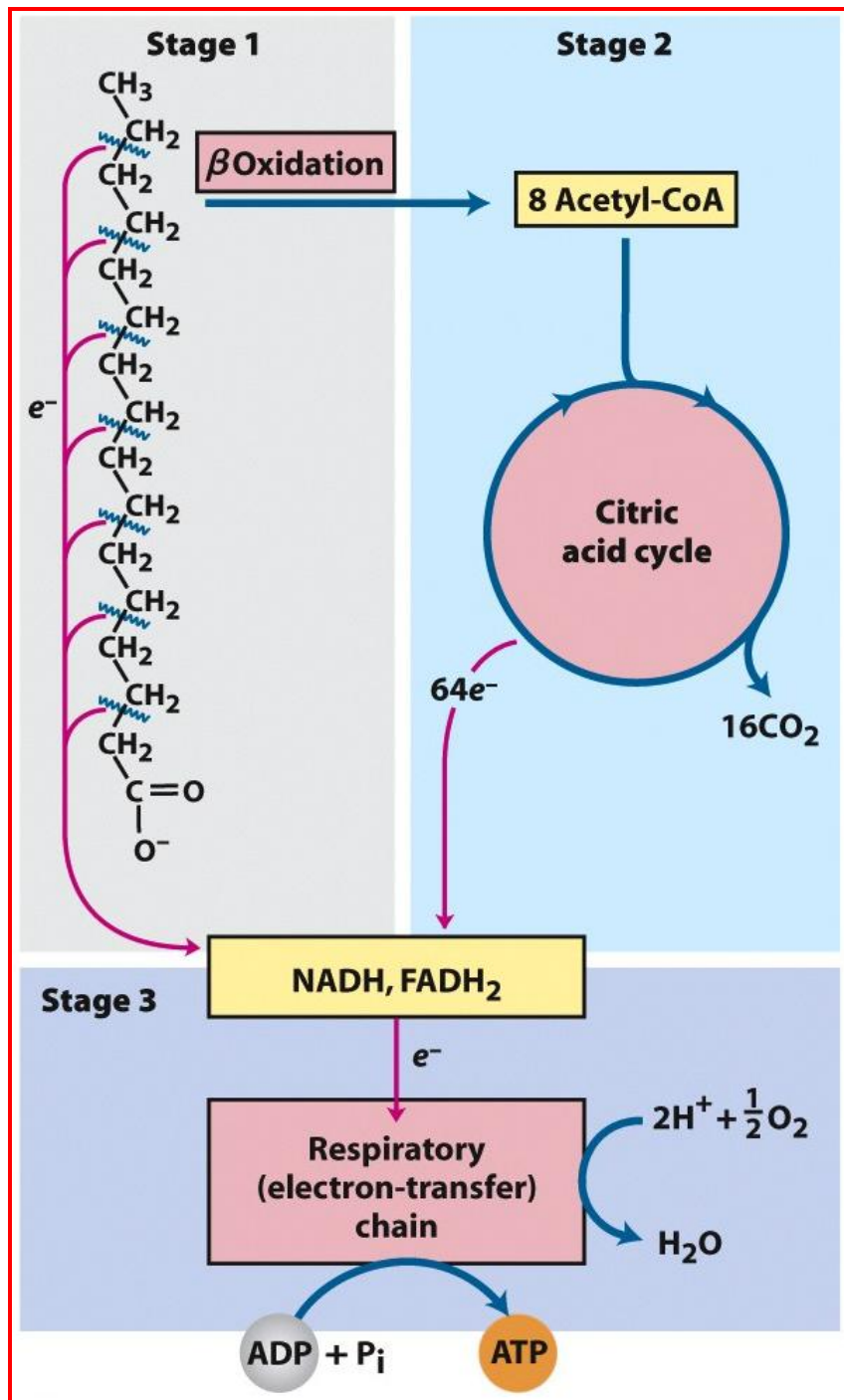
Oxidative degradation



17.2 Oxidation of fatty acids



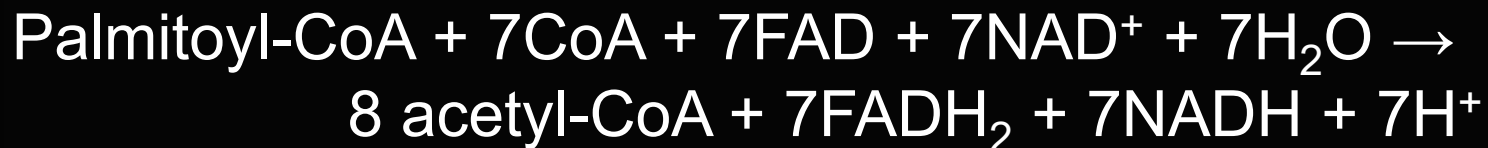
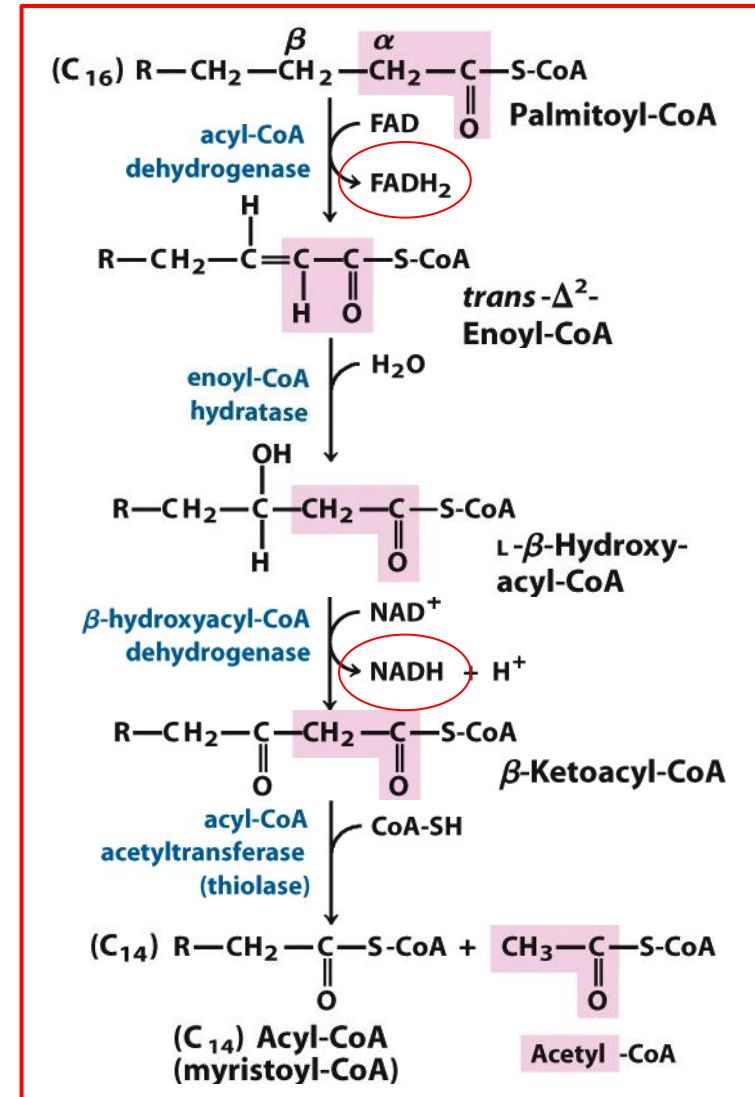
Stages of fatty acid oxidation



1) β -oxidation of saturated FA's

– 4 basic steps

- ① Dehydrogenation of C_n acyl-CoA
- ② Water addition
- ③ Dehydrogenation: **specific for L isomer**
- ④ Thiolysis (cleavage): **C_{n-2} acyl-CoA + Acetyl CoA.**



Acyl-CoA dehydrogenase : 3 isozymes

VLCAD (12-18 carbons), MCAD (4-14), SCAD (4-8)

Electron-transferring flavoprotein (ETF) w/ FAD

steps ② – ④: two sets of enzymes

(1) Multifunctional complex : **trifunctional protein (TFP)**

(2) A set of 4 soluble enzymes



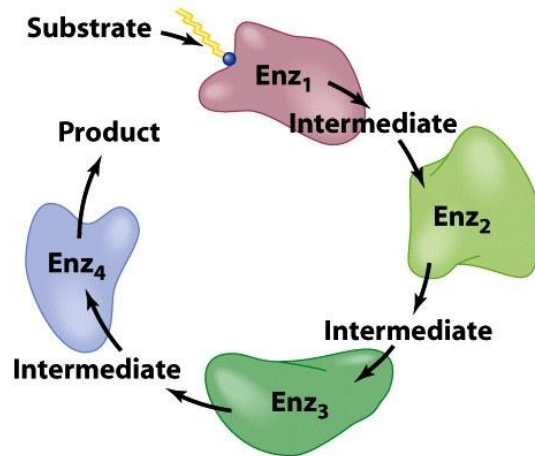
12 or fewer carbons FA
Mitochondrial matrix



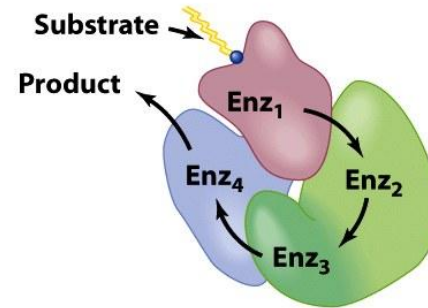
12 or more carbons FA
Inner mitochondrial membrane
(substrate channeling)

Multifunctional protein complex for β -oxidation

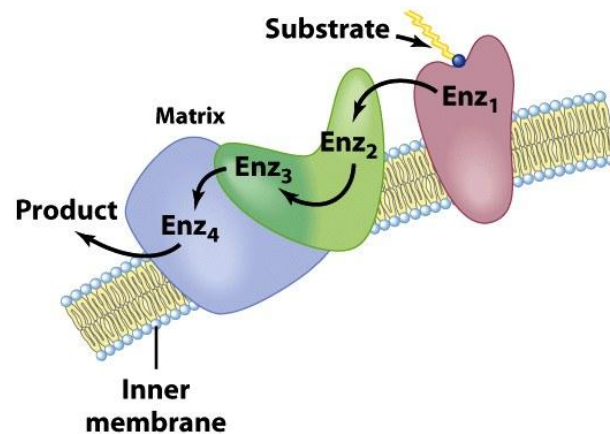
(a) Gram-positive bacteria and mitochondrial short-chain-specific system



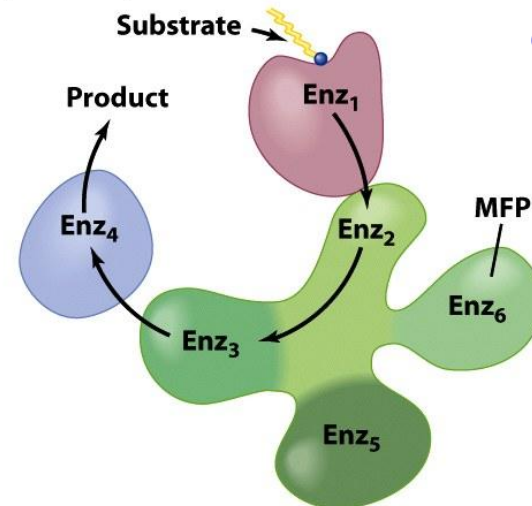
(b) Gram-negative bacteria



(c) Mitochondrial very-long-chain-specific system

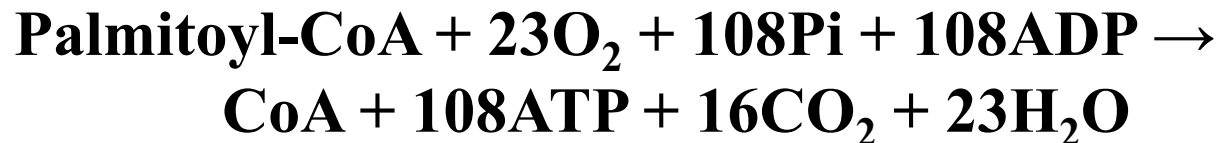
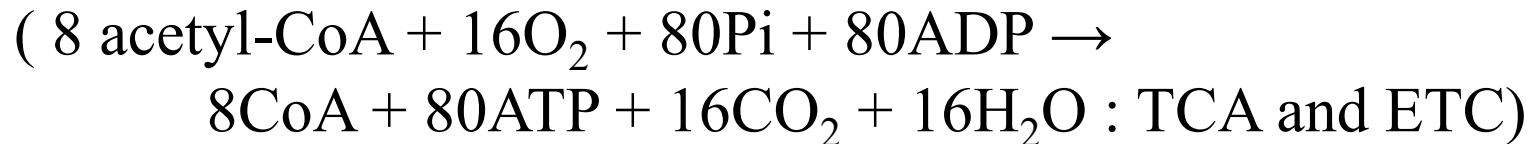
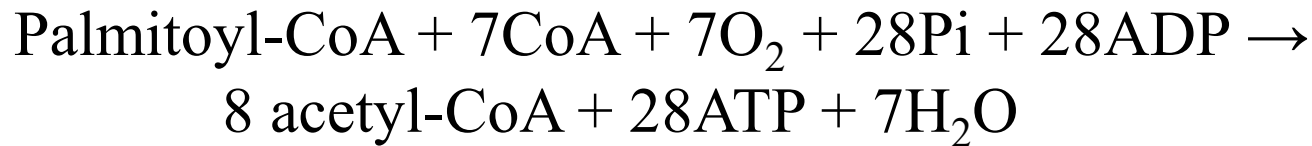
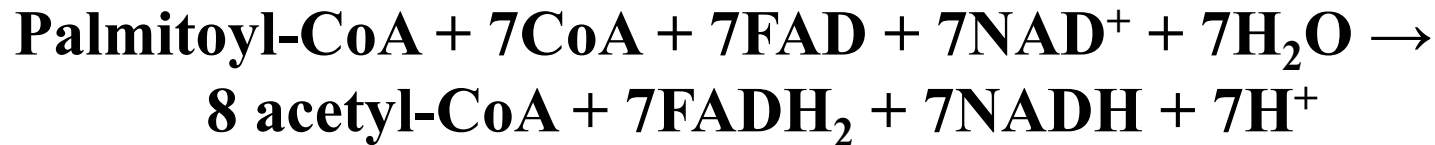
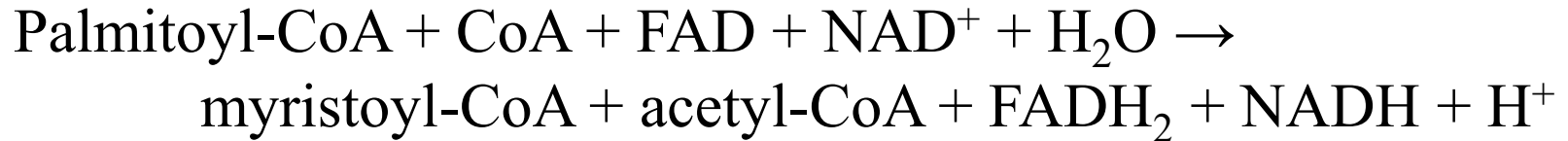


(d) Peroxisomal and glyoxysomal system of plants



Advantage & disadvantage of MFP

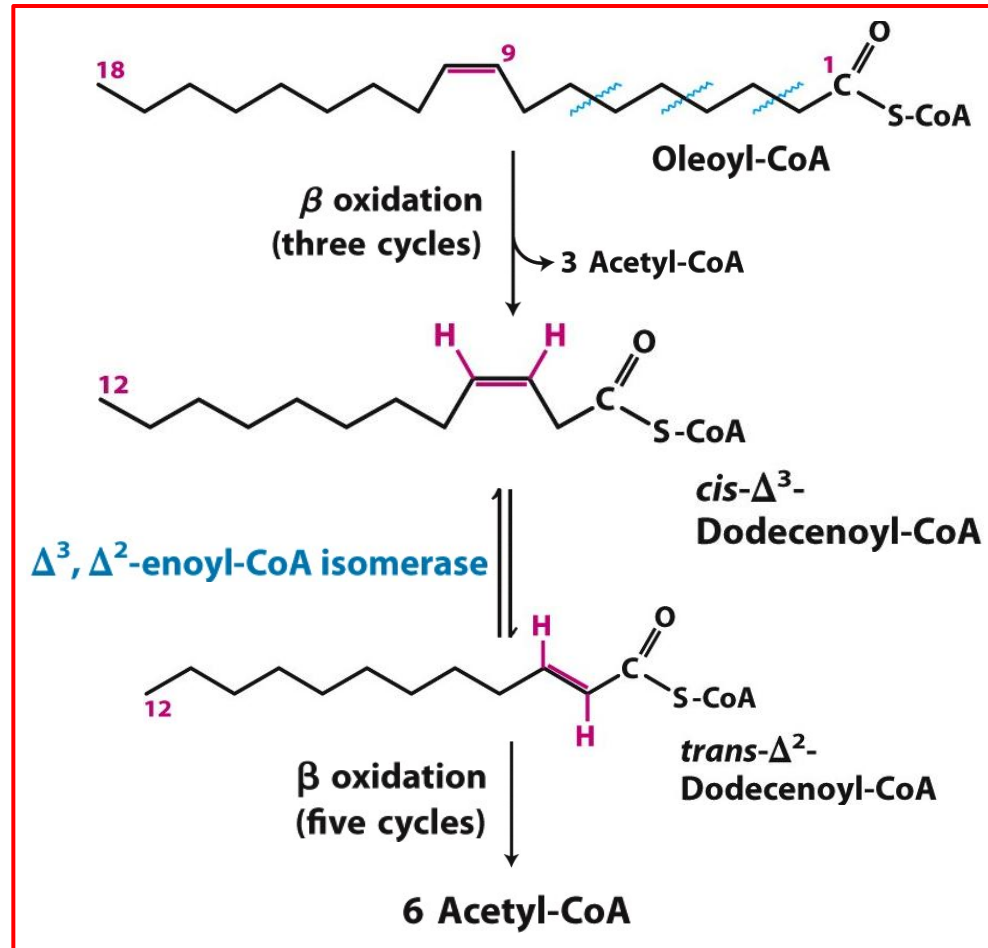
The balance sheet for the oxidation of fatty acids



2) β -oxidation of unsaturated FA's

– An additional step

① **Isomerase**
(trans- Δ^2 -enoyl-CoA)



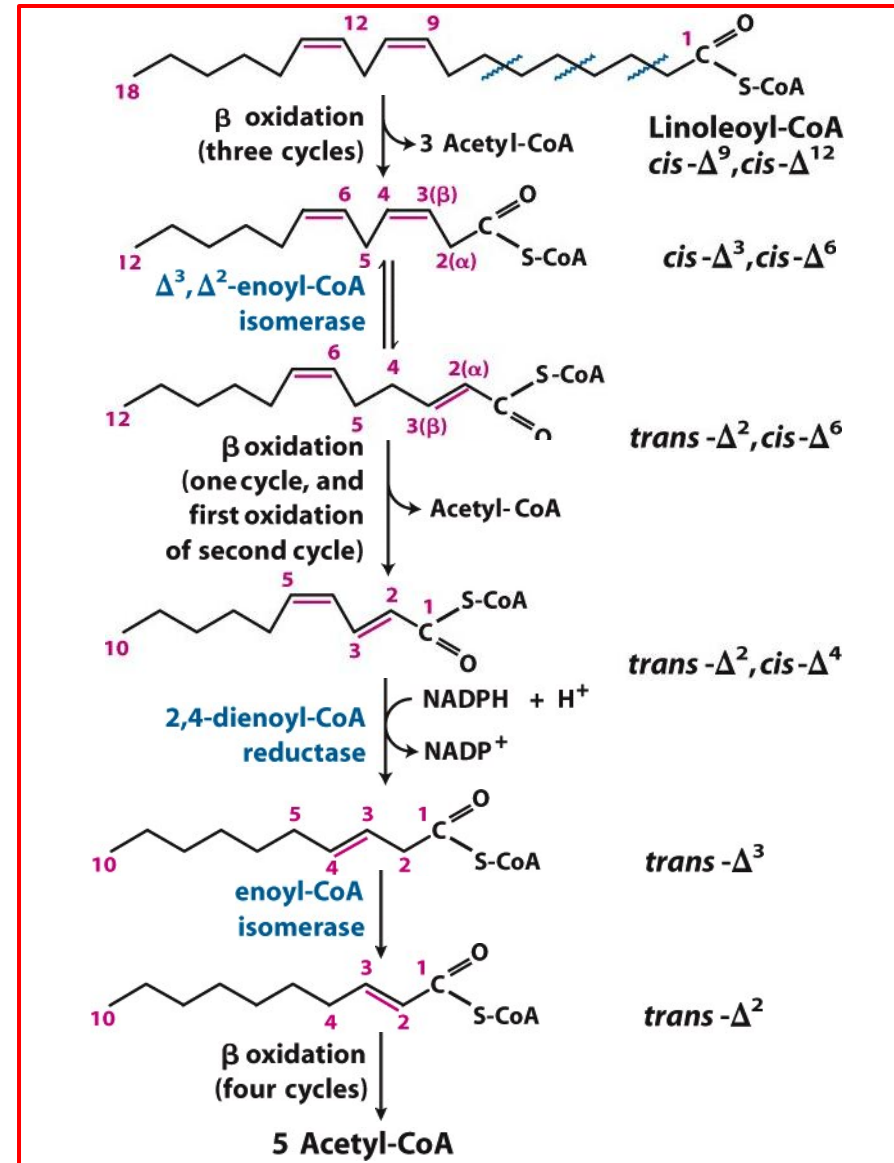
2) β -oxidation of polyunsaturated FA's

– 2 additional steps

① Isomerase

② Reductase

(trans- Δ^2 -enoyl-CoA)



3) β -oxidation of odd-number FA's

- Long chain odd-number FA's will be oxidized as even-number FA's;

finally propionyl CoA is left

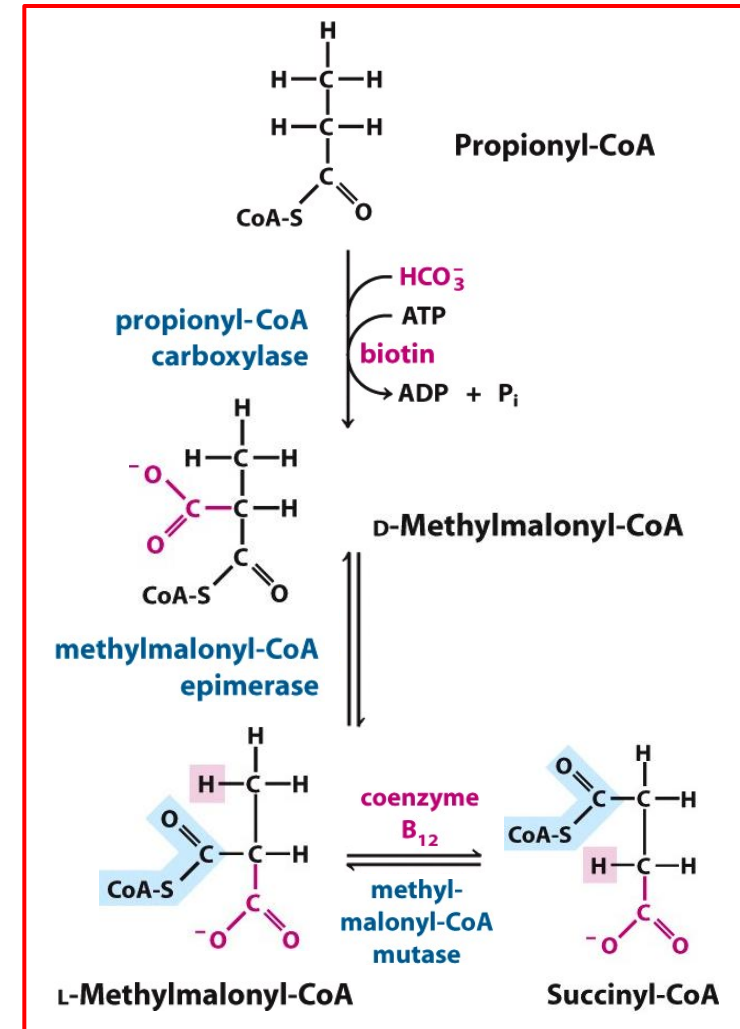
- 3 additional steps

① Carboxylase: **addition of CO_2**

② Epimerase: **D- to L-isomer**

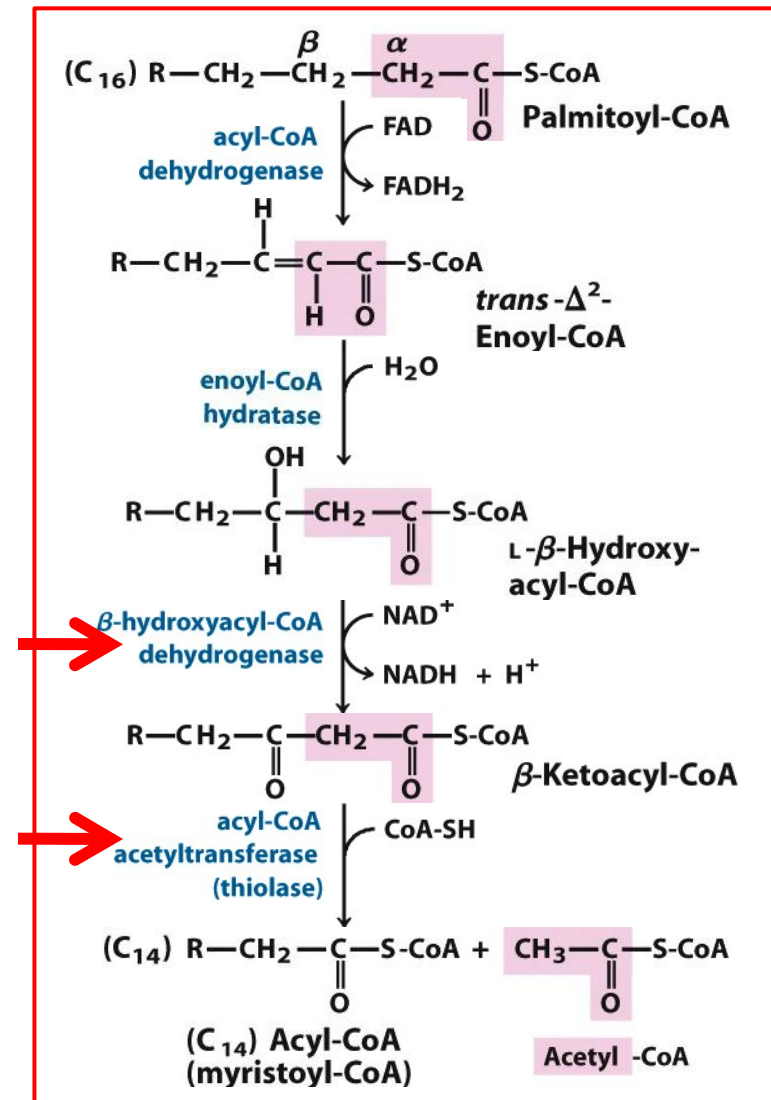
③ Mutase

- **Succinyl CoA can enter citric acid cycle.**



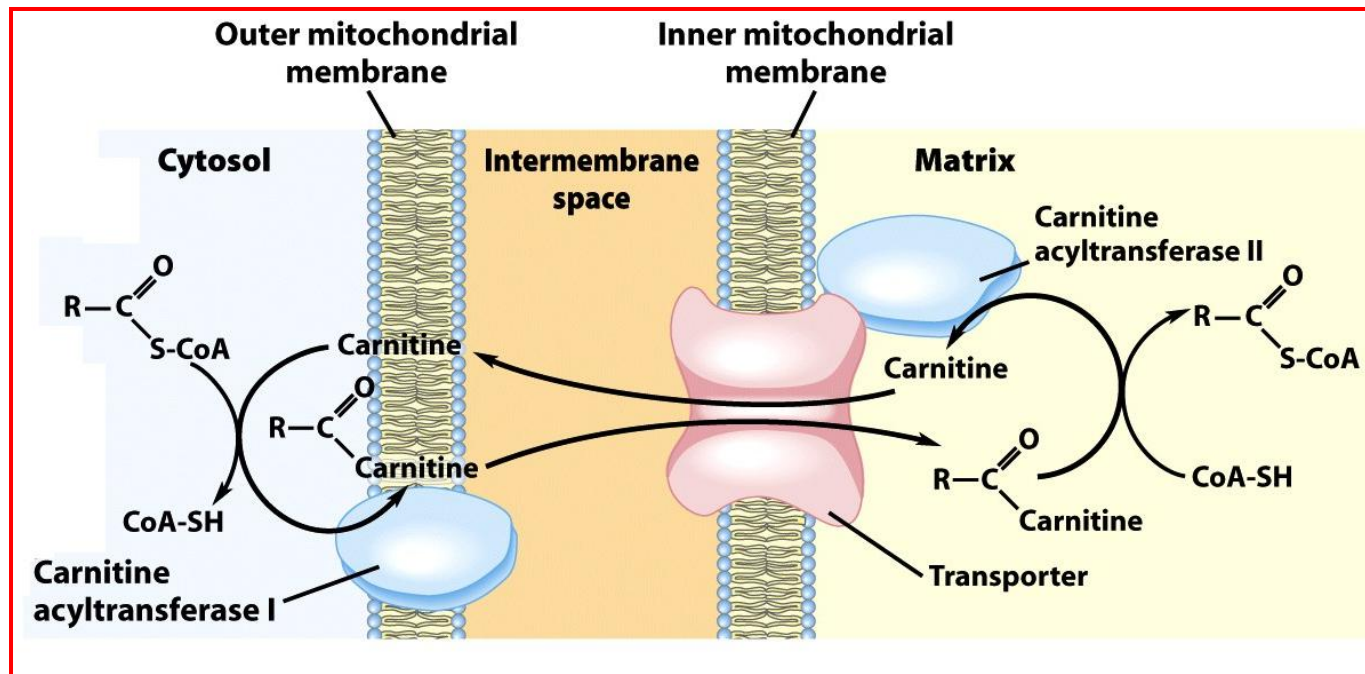
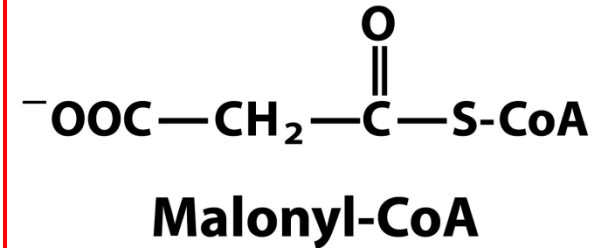
4) Regulation of FA oxidation

- High [NADH]/[NAD⁺] ratio: **inhibits β -hydroxyacyl-CoA dehydrogenase**
- High [acetyl-CoA]: **inhibits thiolase.**

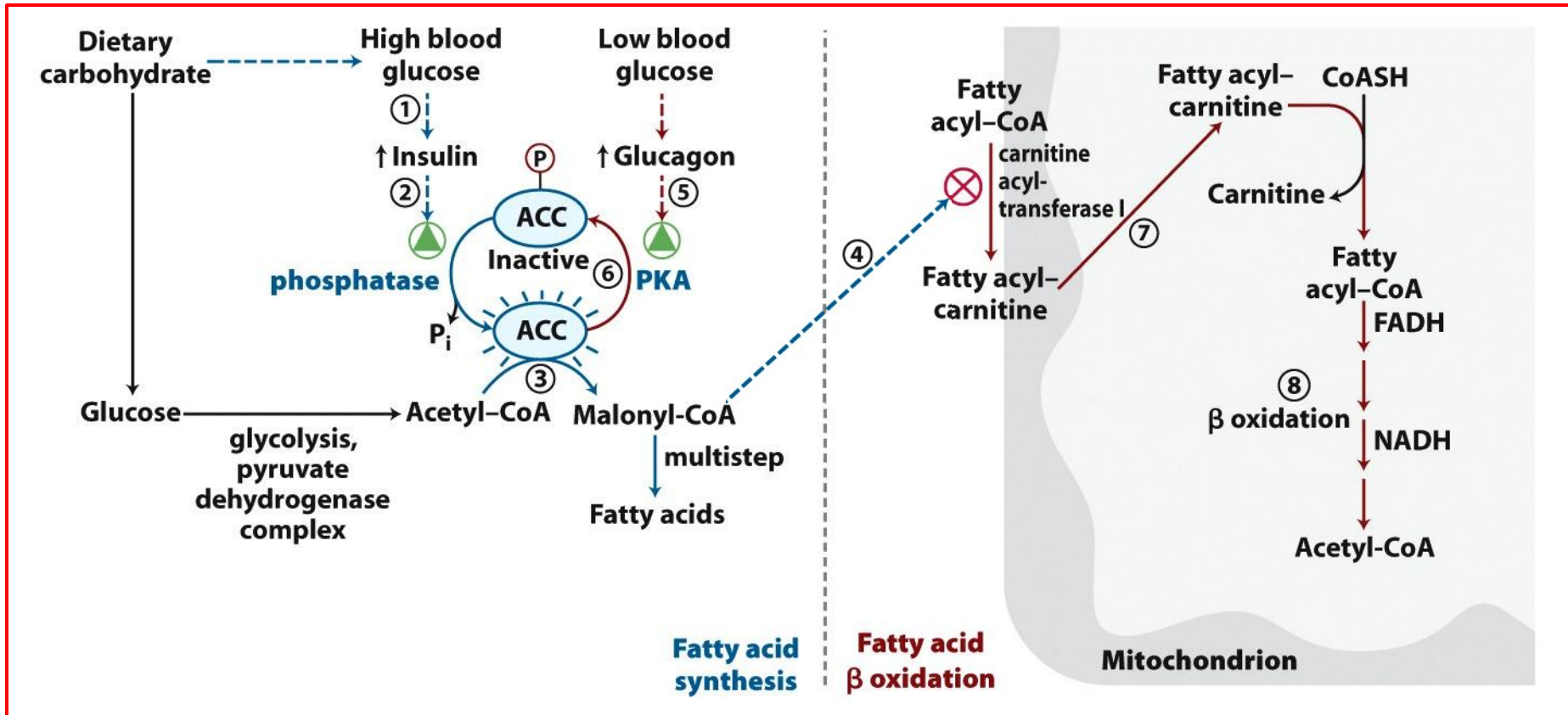


4) Regulation of FA oxidation

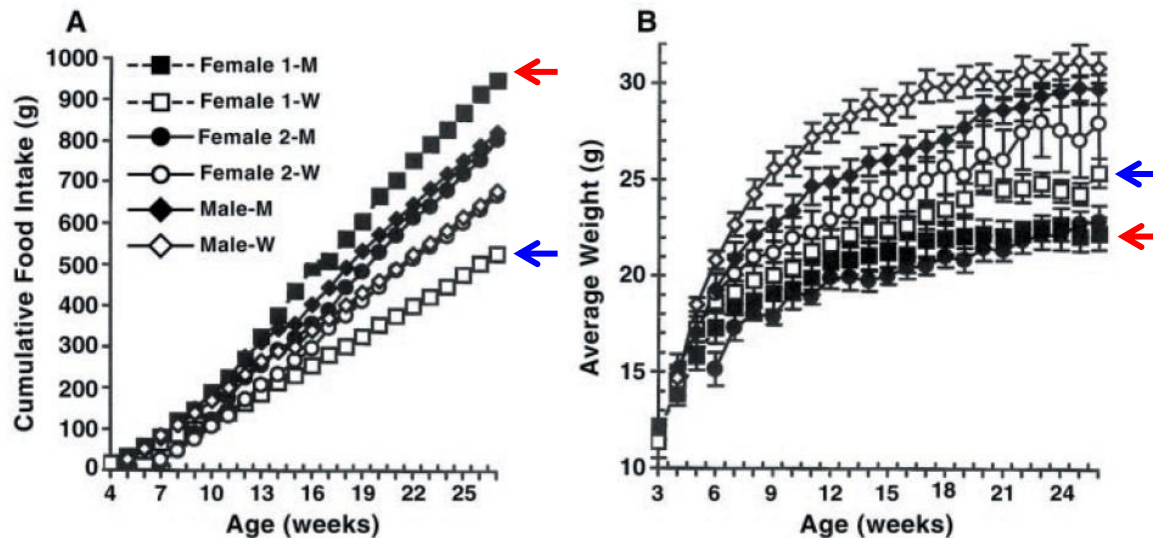
- Malonyl CoA: **inhibits carnitine acyltransferase I (well-fed condition)**



4) Regulation of FA oxidation malonyl-CoA



ACC2 is a promising drug target for obesity



Acc2 mutation

- normal life span
- higher fatty acid oxidation rate
- lower amount of fat

+/+

-/-

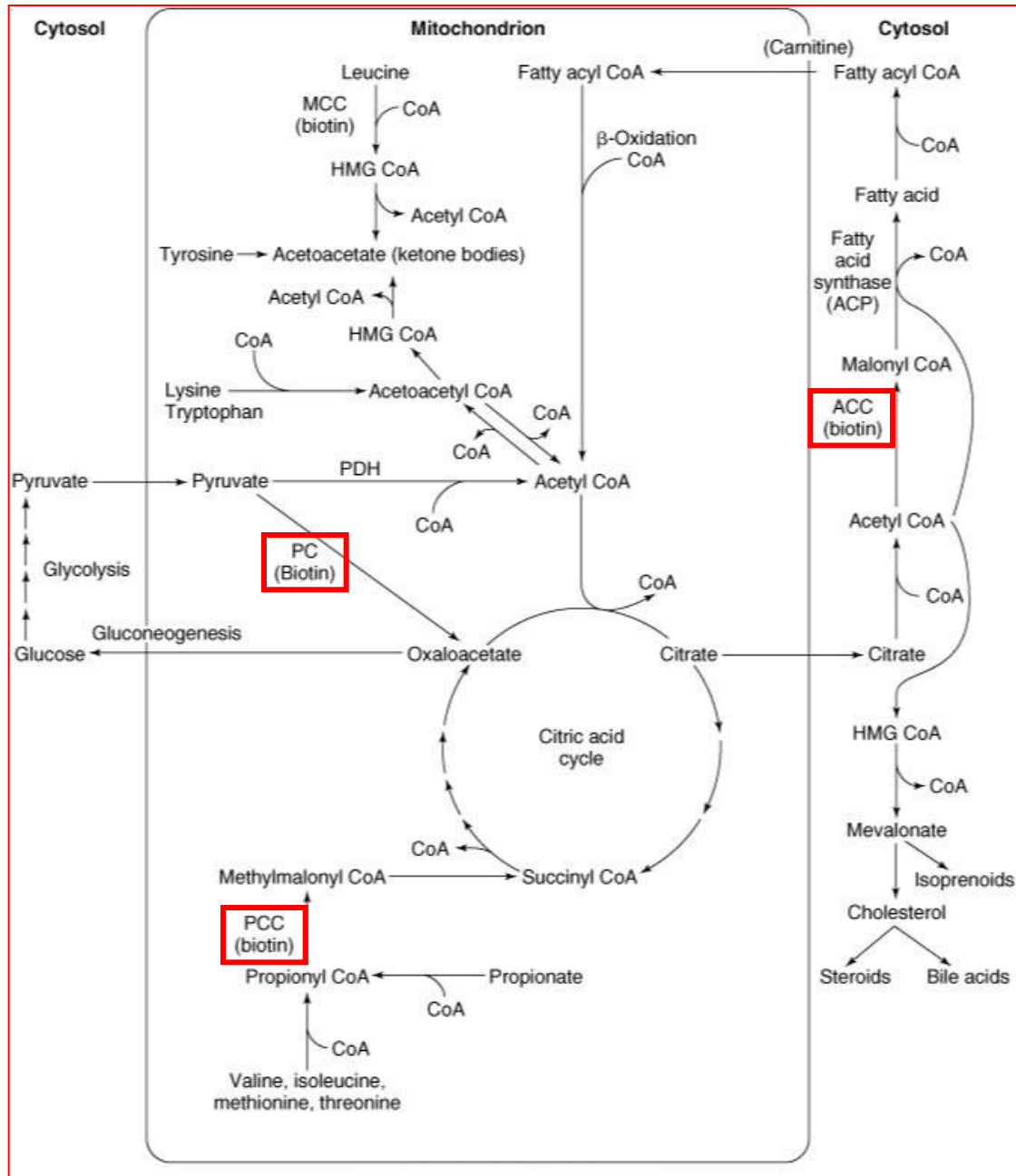


-/-

+/+

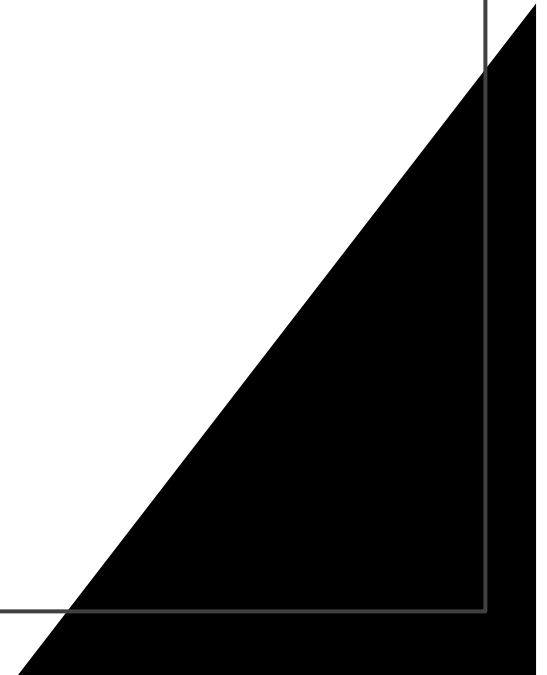


Biotin containing carboxylase

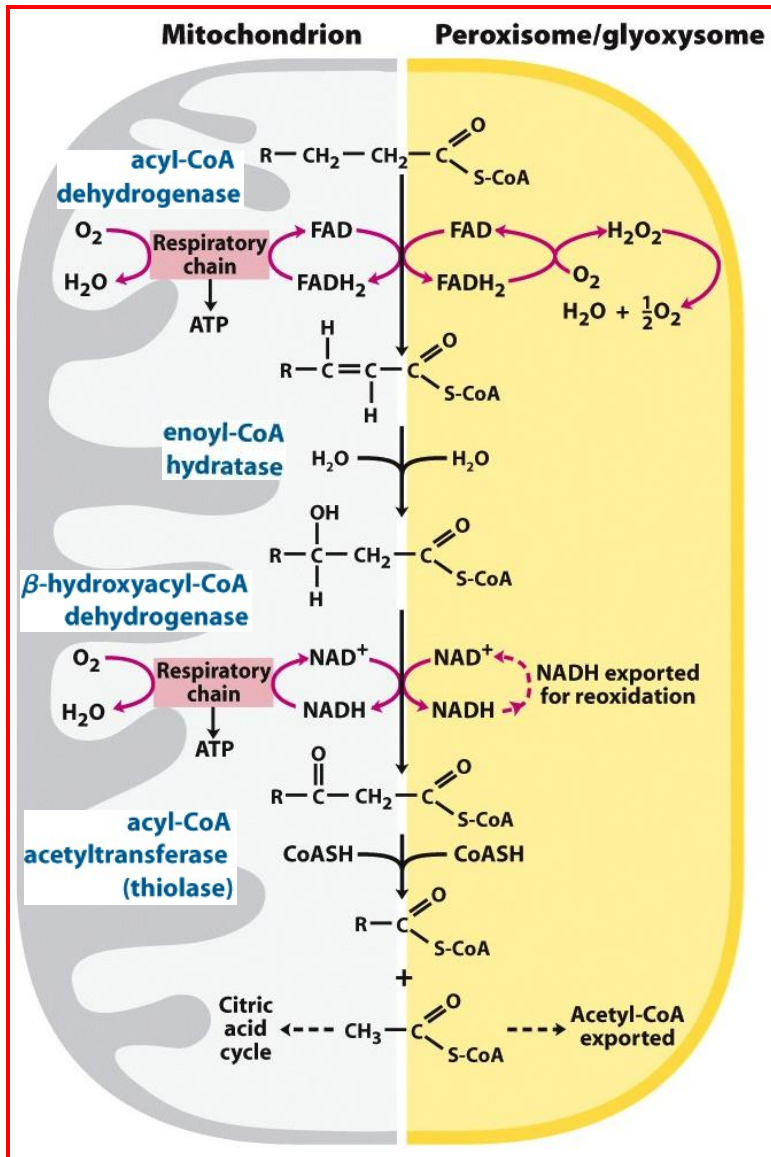
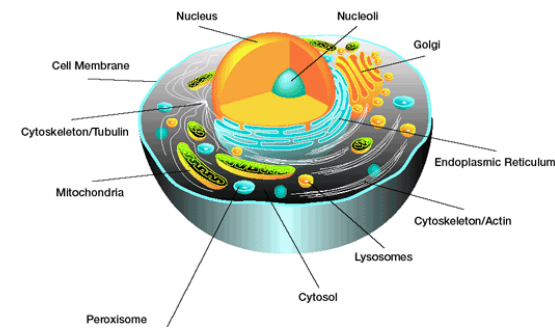


Today, in this class...

- Beta oxidation in peroxisomes
- Ketone bodies
- Electron transfer reactions & photophosphorylation



5) β -oxidation in peroxisome/glyox



– Peroxisome: memb-enclosed organelle of animals and plants

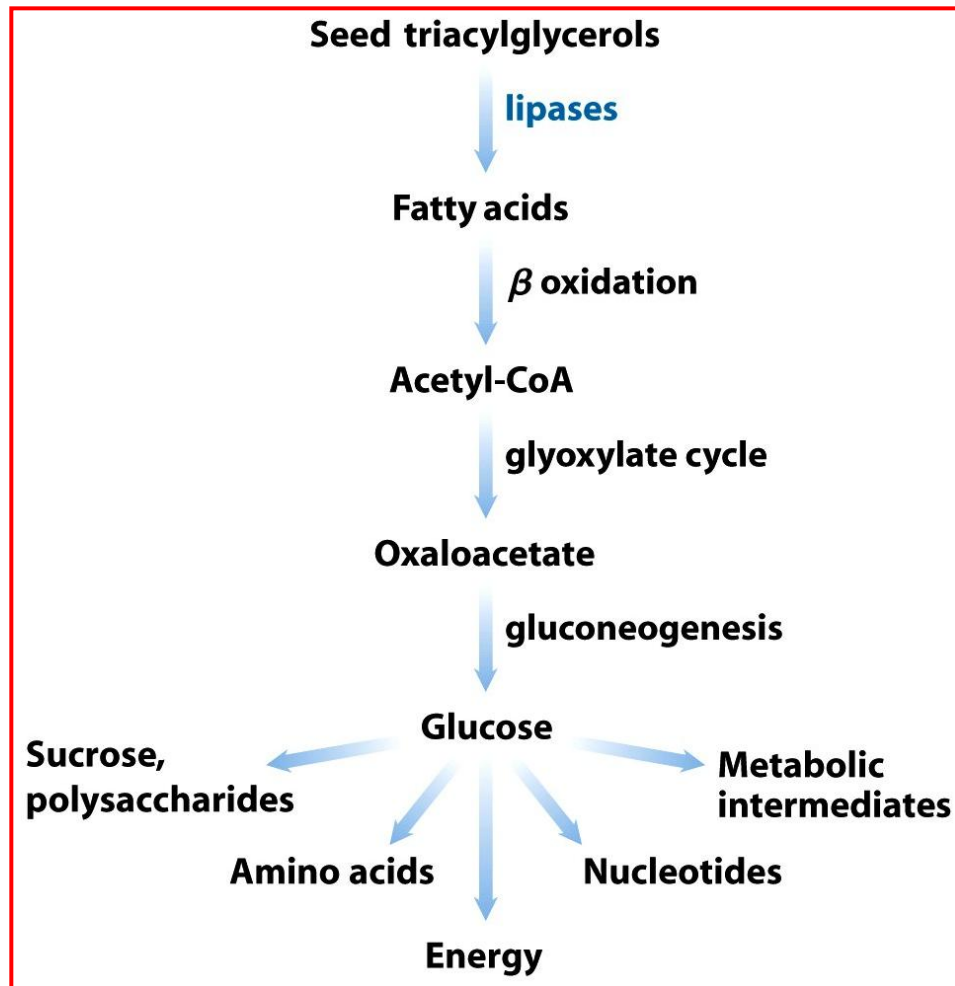
lack of TCA cycle enzymes

- Using H_2O_2 , peroxisomes detoxify various toxic substances that enter the blood (phenols, formic acid, formaldehyde, and alcohol)
- Acetyl-CoA exported for biosynthesis
- NADH exported for reoxidation
- LCFA & BFA $\rightarrow$ short chain FA

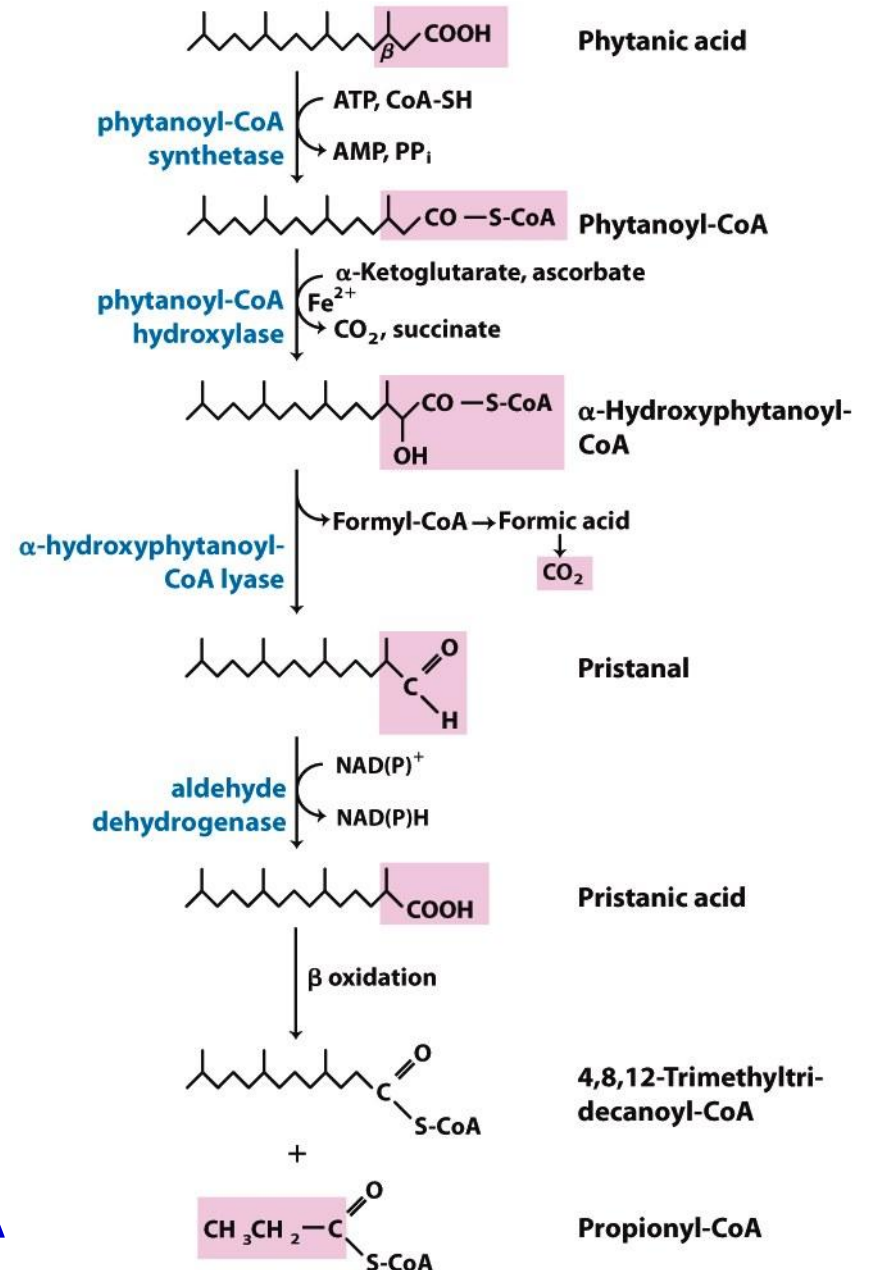
: Zellweger syndrome, X-linked adrenoleukodystrophy (XALD)

β -oxidation in peroxisome/glyoxysome

- Plant peroxisomes and glyoxysomes use Acetyl CoA from β -oxidation as a biosynthetic precursor (ex, in germinating seed)



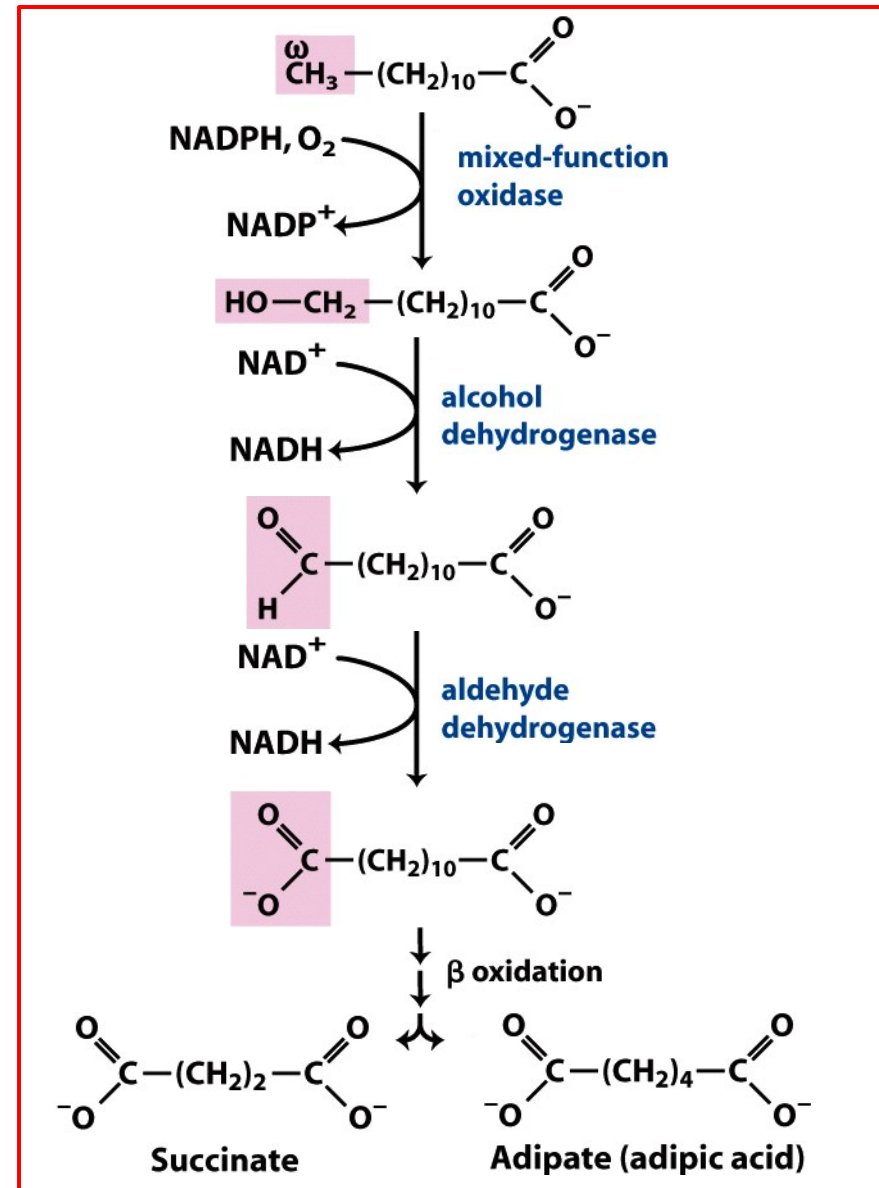
The α oxidation of a branched-chain fatty acid (phytanic acid) in peroxisomes



Synthesis of propionyl-CoA

6) ω -oxidation in ER

- Introduction of a OH grp onto ω C
- O_2 is from cytochrome P450 and electron donor (NADPH)
- After dehydrogenation, either end can attach CoA.

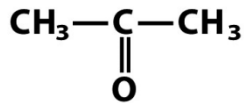


17.3 Ketone Bodies



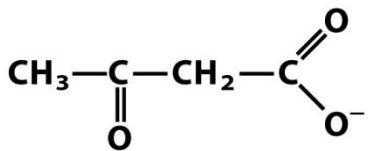
1) Ketone body formation

- Formed in liver and exported to other organs as fuel
- Ketone bodies

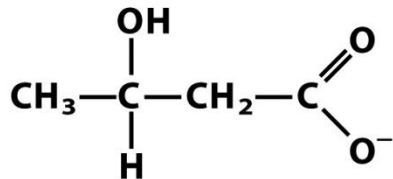


Acetone

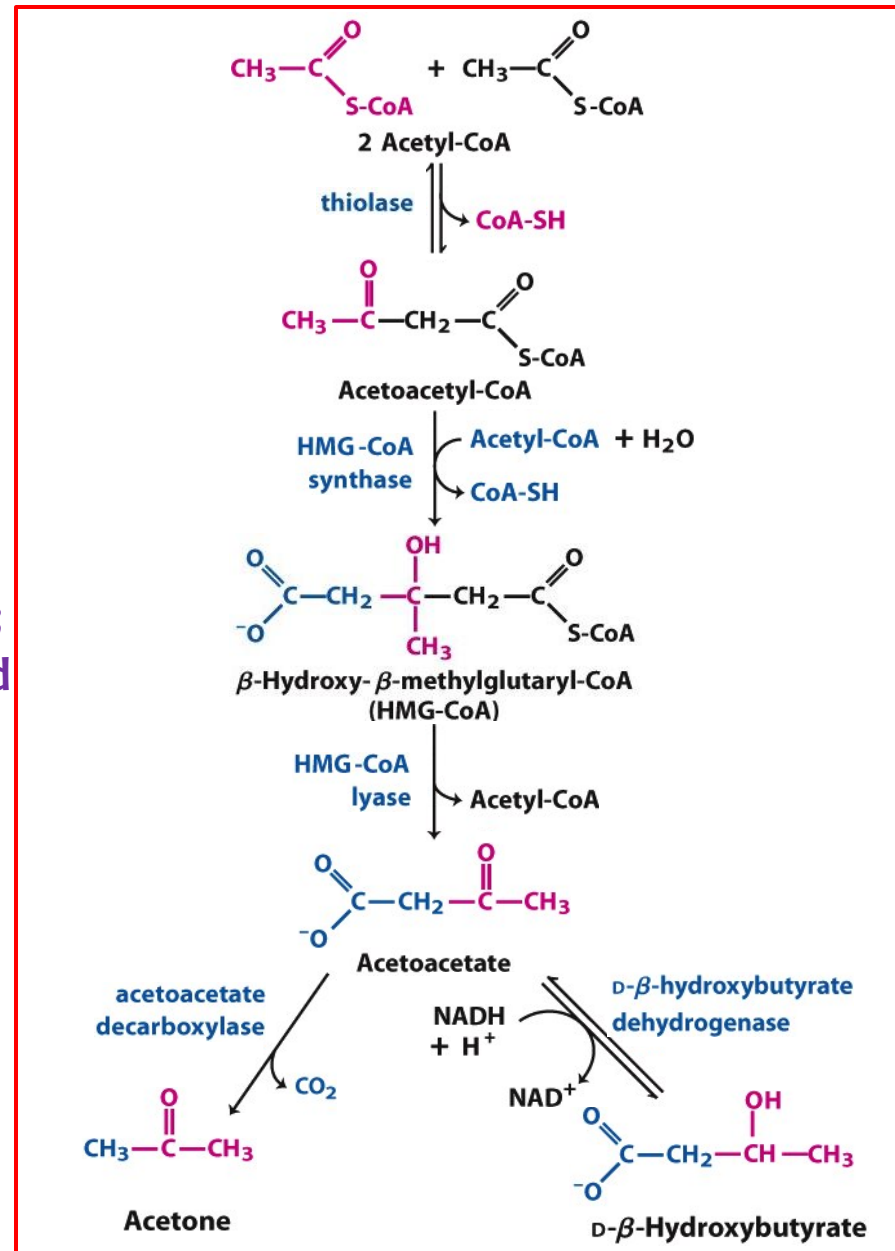
small amount produced;
toxic in blood; is exhaled



Acetoacetate

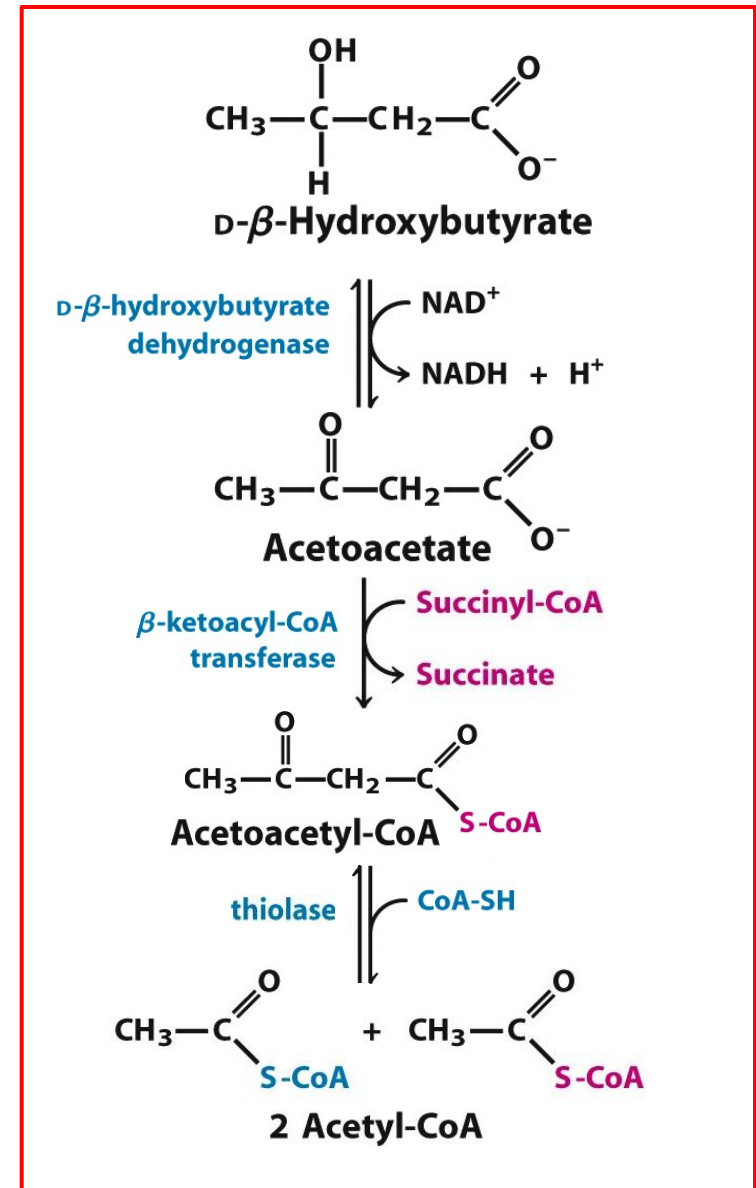


D-β-Hydroxybutyrate

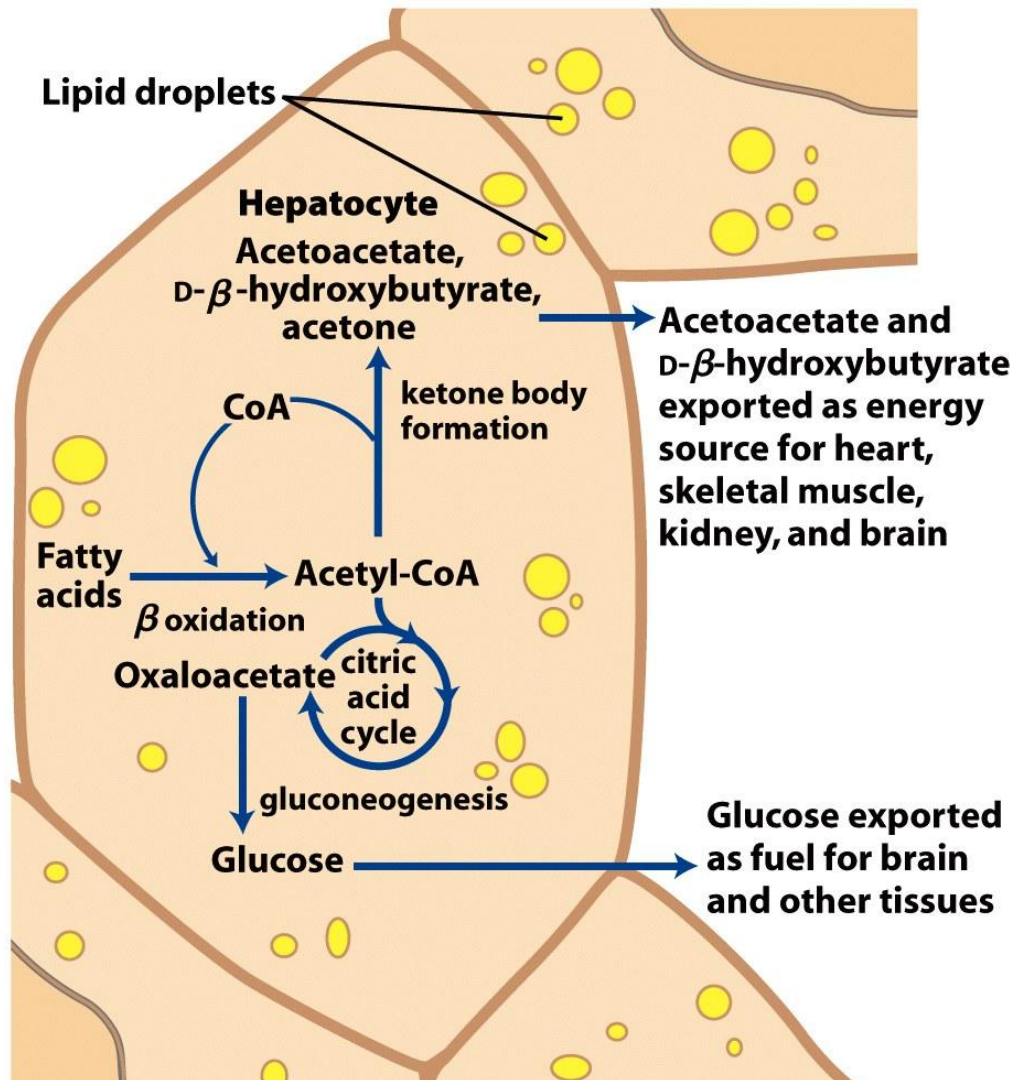


2) Use of ketone bodies

- **D-β-Hydroxybutyrate** and **acetoacetate** can be used as a fuel.
- Happens in extrahepatic tissues: there is no **β-ketoacyl-CoA transferase** in liver.



3) Ketone body formation and export from the liver



Fasting or diabetes



Gluconeogenesis $\uparrow$

Citric acid cycle $\downarrow$

β -oxidation $\uparrow$



Ketone body increased



**Resulting in acidosis,
coma & finally death**

Fluid replacement
Bicarbonate

저탄고지

2020년 한국인 영양소 섭취 기준에 따르면 각 영양소 별 에너지
적정 비율은 성인 기준 탄수화물 55~65%, 단백질 7~20%, 지방
15~30%을 권장하고 있다.

출처 :

https://health.chosun.com/site/data/html_dir/2022/02/09/2022020901793.html

"전체칼로리의 10~25%로 탄수화물을 제한하는 식사의 경우 체중
감소에는 뚜렷한 효과가 없었지만, 혈당과 혈중 지질에는 긍정적인
결과를 보였다"고 말했다. 반면 탄수화물 섭취가 10% 미만인 '아주
낮은 그룹'에서는 6개월 이상이 지나면 오히려 체지방이 증가했고,
나쁜 콜레스테롤인 LDL 콜레스테롤 수치는 상승하는 등 부정적인
결과가 나왔다.

출처 :

https://health.chosun.com/site/data/html_dir/2021/11/11/2021111101762.html

Chapter 17: Summary

In this chapter, we learned:

- Fats are an important **energy source** in animals
- Two-carbon units in fatty acids are oxidized in a four-step **β -oxidation** process into acetyl-CoA
- In the process, a lot of **NADH** and **FADH₂** forms; these can yield a lot of ATP in the electron-transport chain
- Acetyl-CoA formed in the liver can be either **oxidized via the citric acid cycle** or **converted to ketone bodies** that serve as fuels for other tissues